JOURNAL OF PETROLOGY | VOLUME 38 | NUMBER 10 | PAGES 1279–1305 | 1997

Grenville Magmatism in West Texas: Petrology and Geochemistry of the Red Bluff Granitic Suite

WILLIAM M. SHANNON[1]*, CALVIN G. BARNES[1]† AND M. E. BICKFORD[2]

[1]DEPARTMENT OF GEOSCIENCES, TEXAS TECH UNIVERSITY, LUBBOCK, TX 79409-1053, USA

[2]DEPARTMENT OF EARTH SCIENCES, SYRACUSE UNIVERSITY, SYRACUSE, NY 13244-1070, USA

RECEIVED SEPTEMBER 30, 1995; REVISED TYPESCRIPT ACCEPTED MAY 14, 1997

The Red Bluff granitic suite (RBGS) underlies part of the Franklin Mountains in west Texas. It was emplaced into a Middle Proterozoic shelf sequence at 1120±35 Ma. The suite is predominantly metaluminous; it consists of early granitic to quartz syenitic sills, a main mass of alkali feldspar granite, small bodies of alkali feldspar syenite, leucogranitic dikes, and late-stage peralkaline arfvedsonite granite. Late-stage, mildly alkaline (transitional) ferrobasaltic dikes are also present. The suite shows continuous compositional variation from quartz syenite to leucogranite. It is characterized by an iron-rich mineral assemblage, estimated $f(O_2)$ *near FMQ, initial emplacement temperatures near 1050°C, and high concentrations of high field strength and rare earth elements. These features serve to classify the suite as A-type; trace element discrimination diagrams identify it as 'within plate'. When plotted in multielement diagrams, trace element patterns of both the granitic rocks and the ferrobasalts are generally subparallel to those of ocean island basalts. No elemental evidence for a subduction-related origin is present among the granitic rocks. Major element mass-balance models that use observed phases are consistent with an origin by fractional crystallization from a transitional ferrobasaltic parent. However, trace element tests of these models indicate that trapped liquid remained in the crystalline residue ('in situ crystallization'). Partial melting models that use known basement rocks as parental compositions generally fail to reproduce the observed trace element patterns, and those that fit individual rock compositions cannot explain the continuous compositional variation of the suite. Derivation from basaltic parental magmas indicates that the RBGS represents addition of juvenile material to the crust during Grenville time. The mildly alkaline nature of the granitic suite and its lack of subduction-related geochemical features are consistent with an origin in a zone of regional extension, as suggested by recent geophysical and petrological studies.*

KEY WORDS: *alkaline granite; geochemistry; Grenville; geochronology; Texas*

INTRODUCTION

Grenville age magmatism in Texas is known from widely scattered exposures and subsurface (borehole) data. Limited petrologic study of these rocks has yielded a range of ideas with regard to their tectonic setting and petrogenesis. These include subduction-related activity (Nelson & DePaolo, 1985; Norman *et al.*, 1987; Roths, 1993) and, in west Texas, continental rifting (Keller *et al.*, 1989; Adams & Keller, 1994; Kargi & Barnes, 1995). The latter workers cited the tholeiitic Nellie layered mafic intrusion (1·16–1·07 Ga, U–Pb on zircon; Keller *et al.*, 1989; Fig. 1a) as an example of rift-related magmatism during Grenville time.

Alkaline granitic and rhyolitic rocks with U–Pb (zircon) ages from 1046 to 1171 Ma (Wasserburg *et al.*, 1962; Copeland & Bowring, 1988; Roths, 1993; this work) underlie a wide area of far western Texas. Although the Red Bluff granitic suite (RBGS) is the best exposed and most extensively studied of these units, no consensus

*Present address: 9180 Coors Blvd., NW, Apt. 3606, Albuquerque, NM 87120, USA

†Corresponding author. Telephone: 806-742-3106 or 3102. Fax: 806-742-0100. e-mail: gical@ttacs.ttu.edu

© Oxford University Press 1997

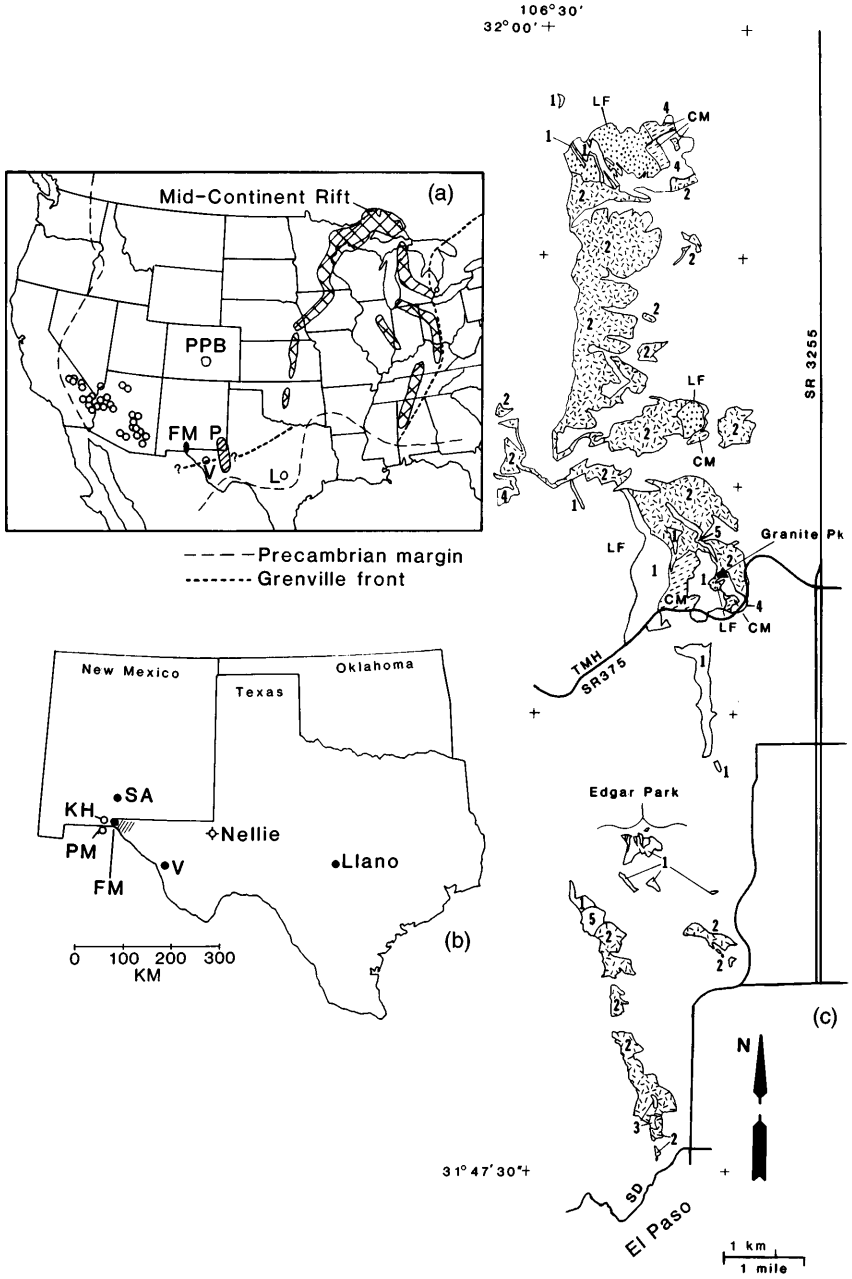


Fig. 1. (a) Selected locations of ~1·1 Ga magmatism in the USA: Mid-Continent Rift, Pikes Peak Batholith (PPB), Pecos mafic intrusive suite (P), Llano Uplift (L), and Franklin Mountains (FM). ○, diabasic dike swarms in the southwestern USA (e.g. Hammond, 1990; Larson *et al.*, 1994). After Adams & Keller (1994). (b) Proterozoic rocks crop out in the Franklin Mountains (FM), Llano uplift, Van Horn area (V), and San Andres Mountains (SA). Cross-hatching indicates area underlain by ~1·1 Ga igneous rocks in west Texas. Granulitic xenoliths are present in maar volcanoes at Kilbourne Hole (KH) and Potrillo maar (PM). Nellie is the location of a well that penetrated layered mafic and ultramafic rocks of the Pecos mafic intrusive complex. (c) Sketch geologic map of the Red Bluff granitic suite. Numbers refer to intrusive stages 1–5 (see text). Wall rocks are the Castner Marble (CM) and Lanoria Formation (LF). The Trans-Mountain highway (SR 375) traverses Fusselman Canyon. After Harbour (1972).

exists on either its petrogenesis or its tectonic setting. For example, Condie (1982) and Norman *et al.* (1987) viewed the RBGS and associated volcanic rocks as part of a tholeiitic continental back-arc series. Mosher (1993) suggested the suite might be arc related. In contrast, Anderson (1983) classified the RBGS as anorogenic.

A clear understanding of the origin of the RBGS is necessary for any regional synthesis of Grenville

magmatism and tectonism. Its mineralogy, mineral compositions, and geochemistry are unusual in the Proterozoic of North America; similar rocks are present in the Wolf River batholith (Anderson, 1980) and the sodic series of the 1·1 Ga Pikes Peak batholith (Smith *et al.*, 1997). In this contribution, we present field, petrographic, geochemical, and geochronological data that bear on the origin of the suite and, in particular, indicate the absence of subduction-related felsic magmas in the RBGS. Geochemical models are then used to demonstrate that the suite was derived by fractional crystallization from a basaltic parent and thus represents addition of juvenile material to the continental crust during Grenville time.

GEOLOGIC SETTING

The RBGS and associated volcanic rocks are exposed in the Franklin Mountains, a westward-tilted fault block (Harbour, 1972; Lovejoy, 1975) that extends ~30 km northward from El Paso, Texas (Fig. 1). The granitic magmas intruded an undeformed Proterozoic section that consists, from oldest to youngest, of the Castner Marble, Mundy Breccia, Lanoria Formation, and Thunderbird Group (Harbour, 1960; Thomann, 1980; Fig. 2). The first three units comprise a contact metamorphosed shelf sequence of undeformed carbonates and mudrocks, basaltic breccia, and sandstone, respectively (Harbour, 1960; Hoffer, 1976; Anthony *et al.*, 1991; Pittenger *et al.*, 1994). Contact metamorphism was at hornblende hornfels conditions (Hoffer, 1976). U–Pb (zircon) dating of ash-rich beds in the Castner Marble indicates a depositional age of ~1·25 Ga (Pittenger *et al.*, 1994).

The Thunderbird Group consists of trachytic and rhyolitic ignimbrites and lavas, and associated clastic sedimentary rocks (Thomann, 1980). Geochemical evidence (Norman *et al.*, 1987) suggests that the Thunderbird Formation volcanic rocks were comagmatic with the RBGS. Roths (1993) reported ages of 1086 ± 5 Ma for the RBGS and 1111 ± 43 Ma for a rhyolite of the Thunderbird Group.

The basement of the Proterozoic shelf sequence is unknown. Older metamorphic and granitic rocks are exposed to the north in the San Andres Mountains of New Mexico (Fig. 1b; Roths, 1991) and high-grade metamorphic xenoliths are present in the ejecta of the Kilbourne Hole volcano (Padovani & Carter, 1977). The Proterozoic rocks exposed to the north in New Mexico comprise gneiss, amphibolite, quartzite, and undeformed ~1·35 Ga granite (Condie & Budding, 1979; Roths, 1991). Crustal xenoliths from Kilbourne Hole include granulite facies mafic and granodioritic orthogneiss, pelitic gneiss, and anorthosite (Padovani & Carter, 1977).

Intrusive relations

Intrusion of the Red Bluff granitic suite occurred in five stages, each of which consists of compositionally distinct rock types. The intrusive units are described below in order of intrusion, which is illustrated in Fig. 2. Rock names follow the IUGS convention (Streckeisen, 1974).

Stage 1

Stage 1 consists of granitic to quartz syenitic sills in the Lanoria Formation and Castner Marble. The best exposures are in Fusselman Canyon along SR 375 (Fig. 1c), where the structurally higher sill reaches a thickness of 660 m (Ray, 1982). In this location, the sill margins are porphyritic alkali feldspar granite that grades inward to coarsely glomeroporphyritic alkali feldspar quartz syenite (Ray, 1982) with rare equigranular syenitic enclaves. The lateral extent of these sills is obscured by high-angle faults, but a group of meter- to 10-m-thick porphyritic alkali feldspar granite sills is exposed to the south at the same stratigraphic level (Fig. 1c) and is thought to be distal parts of the Fusselman Canyon sills.

In the northern part of the range (Fig. 1c), three sills of porphyritic clinopyroxene–alkali feldspar–quartz syenite crop out. On the basis of intrusive style, total thickness, and chemical composition, these sills are thought to belong to stage 1.

Stage 2

About 80% of the granitic rock exposed in the Franklin Mountains consists of massive, coarse-grained, relatively homogeneous, biotite–amphibole–alkali feldspar granite. This unit cuts stage 1 sills and is therefore referred to as stage 2. It contains rare oblate syenitic enclaves that are compositionally identical to stage 3. In the northern part of the area, massive clinopyroxene–alkali feldspar granite of stage 2 crosscuts the Castner Marble and the Lanoria Formation and lies structurally below the stage 1 sills.

Stage 3

Biotite–amphibole–alkali feldspar–quartz syenite makes up stage 3. It is medium to coarse grained and typically massive. Weak magmatic foliation, as defined by aligned amphibole and feldspar, is rare. The principal outcrop area is in the southern part of the range (Fig. 1c), where the syenite intrudes stage 2 granite. In the north, syenitic sills and dikes cut the wall rocks at the same stratigraphic position as stage 1 sills in the Fusselman Canyon area.

Stage 4

Stage 4 consists of leucocratic biotite–alkali feldspar granite that occurs as dikes and, less commonly, larger, boss-like masses (Fig. 1c). Stage 4 rocks intrude all of the older

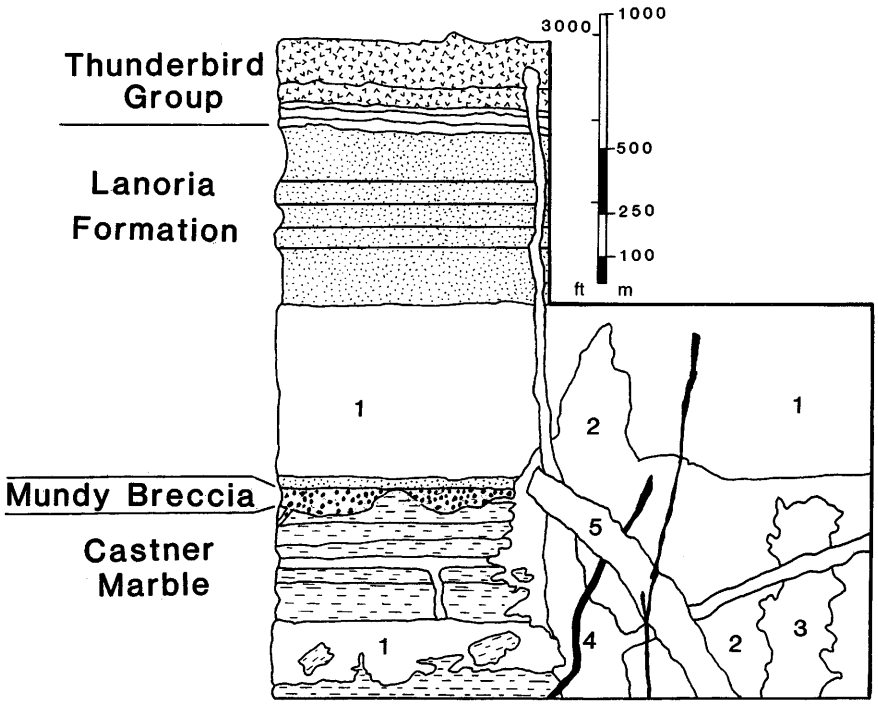


Fig. 2. Diagrammatic wall rock stratigraphic column and the intrusive sequence of the RBGS. Intrusive stages are numbered in order of emplacement: Stage 1 sills range from alkali feldspar quartz syenite to porphyritic alkali feldspar granite. Stage 2 is alkali feldspar granite, stage 3 is alkali feldspar quartz syenite, stage 4 is leucogranite, and stage 5 is arfvedsonite granite, commonly pegmatitic. Ferrobasaltic dikes are shown as thick black lines. After Anthony *et al.* (1991).

intrusive units and also ignimbrites of the Thunderbird Group on the west side of the range.

Approximately 3·7 km north of Fusselman Canyon, leucocratic white mica–topaz–alkali feldspar granite is host to cassiterite-bearing pegmatitic veins. Although mineralogically distinct from most stage 4 rocks, this dike is included with the group because of its association with abundant stage 4 dikes in the area.

Stage 5

The youngest granitic intrusive unit forms dikes of biotite–arfvedsonite–alkali feldspar granite. The dikes are locally banded or layered (i.e. line rock). Their upper zones are commonly pegmatitic, with well-developed comb-layering in which arfvedsonite and alkali feldspar reach 5 cm in length and zircon crystals are commonly 0·5 mm across, and rarely are 20 mm across.

In the southern part of the range, anastomosing stage 5 dikes intrude porphyritic granite of stage 1 in a zone ~300 m wide. Sodium metasomatism is widespread in the host rocks in this zone.

Late-stage ferrobasalts

North-striking, mildly alkaline (transitional) ferrobasaltic dikes (1–10 m wide) intrude the granitic rocks and rocks of the Proterozoic sedimentary section throughout the range. In general, all stages of the granitic suite are cut by the late-stage dikes, but two locations in Fusselman Canyon are noteworthy. At one, a ferrobasaltic dike cuts the Castner Marble and is cut by stage 4 leucogranite. This ferrobasalt may be a feeder for the Mundy Breccia, in which case it would be considerably older than the RBGS (Harbour, 1960). At the other, a ferrobasaltic dike cuts a leucogranite dike of stage 4, but is itself cut by a subhorizontal stage 5 dike. This relationship indicates that during emplacement of the RBGS, basaltic magmas were present at least as early as stage 4.

GEOCHRONOLOGY

A sample of the main granitic phase of the RBGS, stage 2, was processed to separate zircons for age determination. The sample yielded an abundance of zircons with well-developed prismatic forms and terminations (Appendix Fig. A1). Growth zoning is prominent and is consistent with an igneous origin. Examination in high refractive index oils revealed numerous dark inclusions but no inherited cores were observed.

The data (Table 1; Fig. 3) yield an upper intercept age of 1120 ± 35 Ma, with a lower intercept of 144 ± 150

Table 1: U–Pb isotopic data for zircon from Stage 2, granite sample FM-157

Sample*	Concentration†			Pb isotopic composition‡			Radiogenic ratio§			Age (Ma)¶		
FM-157	Wt (mg)	U (p.p.m.)	Pb (p.p.m.)	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{206}Pb$	$^{208}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{207}Pb/^{235}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$
nm(0)	0·2494	28·45	5·20	5615·3	0·07903	0·17958	0·16832	1·7755	0·07651	1003±15	1036±16	1108±2
m(1)sm	0·2418	258·79	41·60	1563·6	0·08466	0·19121	0·14475	1·5081	0·07557	871±13	934±14	1084±7
m(1)lg	1·6682	372·27	55·29	3261·2	0·08010	0·15613	0·13887	1·4503	0·07574	838±13	910±14	1088±1
m(2)lg	0·6245	378·35	59·80	3896·8	0·07951	0·16126	0·14739	1·5417	0·07586	886±13	947±14	1091±1

*nm, nonmagnetic; m, magnetic; numbers in parentheses indicate tilt on Frantz separator at 1·7 A; lg, large; sm, small.
†Total U and Pb, corrected for analytical blank.
‡Measured ratios, corrected for blank and 0·1% per a.m.u. mass discrimination.
§Ratios corrected for analytical blank and non-radiogenic Pb [Stacey & Kramers (1975) model].
¶Errors given at 2σ.

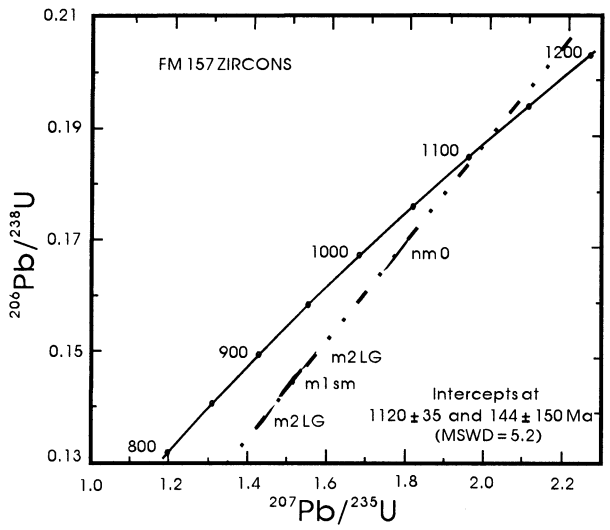


Fig. 3. Discordia diagram of zircon U–Pb isotopic data from stage 2 granite sample FM 157 (Table 1). Ages along concordia in Ma.

Ma and no indication of older, inherited zircons. The relatively large uncertainty in the upper intercept age results from the fact that, despite heavy abrasion, the zircon fractions yielded rather discordant data, probably because of Pb loss. The age obtained thus indicates that the RBGS is coeval with the 1111±48 Ma Thunderbird Group rhyolites (Roths, 1993), and is consistent with geochemical data that indicate that the granites and rhyolites are comagmatic.

PETROGRAPHY

Stage 1

Syenitic (central) parts of stage 1 sills are glomeroporphyritic with hypidiomorphic granular groundmass. Glomerocrysts of perthitic alkali feldspar reach 3 cm in diameter and quartz phenocrysts reach 3 mm in diameter. Clinopyroxene is present as anhedral inclusions in alkali feldspar and amphibole. Amphibole, biotite, quartz, and alkali feldspar are interstitial. Zircon is a euhedral groundmass mineral and, along with ilmenite and fluorite, forms inclusions in amphibole and biotite. Stubby to elongate apatite inclusions are present in the ferromagnesian silicates.

Granitic parts of stage 1 (sill margins and thin sills) are porphyritic, with phenocrysts of alkali feldspar as much as 10 mm in diameter, and of euhedral to skeletal quartz. In the northern exposure of stage 1, some alkali feldspar phenocrysts are cored by plagioclase. Mafic minerals are clinopyroxene, amphibole, and biotite, which form aggregates commonly intergrown with fluorite and subhedral oxide minerals. Clinopyroxene is also sparsely present as microphenocrysts. Accessory zircon ranges from euhedral microphenocrysts as much as 0·5 mm in diameter to interstitial crystals. Prismatic to acicular apatite forms inclusions in the ferromagnesian silicates.

Metasomatized stage 1 rocks contain quartz and alkali feldspar glomerocrysts intergrown with arfvedsonite, astrophyllite, fluorite, and zircon. The fine-grained matrix contains astrophyllite, and arfvedsonite, with accessory fluorite, euhedral to poikilitic zircon, and rare oxides. Arfvedsonite also fills cracks in alkali feldspar, surrounds euhedral zircon and forms rare spherulitic clusters that apparently replace mafic silicates.

Stage 2

Stage 2 granite is coarse grained and hypidiomorphic granular. Perthitic alkali feldspar contains rare cores of subhedral plagioclase as calcic as An_{36}, and exsolved inclusions of albite (An_{0-3}). Except in the north, feldspar-to-feldspar contacts are marked by exsolved albitic plagioclase. Interstices are occupied by quartz, albitic

plagioclase, biotite, fluorite, zircon, and subhedral amphibole that is intergrown with all of the interstitial minerals. Interstitial olive to emerald green clinopyroxene is present in the northern part of the range.

Accessory apatite is present as prismatic inclusions in amphibole. Ilmenite and sparse magnetite form subhedral inclusions in amphibole and in the groundmass. The ratio of magnetite to ilmenite increases as the proportion of biotite to amphibole increases.

Secondary grunerite (after fayalite?) is present as cores in amphibole and as fibrous bundles intergrown with amphibole and biotite. In the northern part of the area, secondary calcite is intergrown with clinopyroxene, and the rock displays sparse miarolitic cavities that are lined with euhedral quartz and albite and filled with calcite.

Stage 3

Stage 3 quartz syenite varies from porphyritic to massive, with coarse grain size and hypidiomorphic granular texture. Sodic orthoclase phenocrysts commonly contain oligoclase cores ($An_{23–31}$). Interstitial and accessory minerals are similar to those in stage 2 samples.

Stage 3 dikes contain feldspar phenocrysts as in the coarse-grained samples. Grunerite (after fayalite?) reaches 1 mm in length, amphibole and biotite poikilitically enclose alkali feldspar and prismatic apatite, and zircon is present as euhedral prismatic microphenocrysts (0·1–0·5 mm long). The groundmass consists of alkali feldspar; elongate groundmass zircon is <0·1 mm long.

Stage 4

Stage 4 granite is hypidiomorphic granular to granophyric, with rare phenocrysts of embayed quartz and perthitic alkali feldspar. Biotite, fluorite, ilmenite, and magnetite are interstitial.

The leucocratic topaz-bearing stage 4 dike is porphyritic with phenocrysts of perthitic alkali feldspar in a fine-grained, hypidiomorphic granular matrix. The rock contains subhedral oikocrysts of quartz, interstitial perthite, pale brown mica, subhedral primary topaz, and rare fluorite. Zircon, ilmenite, and magnetite are included in the alkali feldspar.

Stage 5

The texture of stage 5 granitic dikes ranges from xenomorphic granular to pegmatitic. Quartz, biotite, arfvedsonite, and zircon are commonly poikilitic around fluorite, albite, and microcline. The quartz and arfvedsonite typically are euhedral against each other. The principal Ti-bearing mineral is astrophyllite, which occurs as phenocrysts and interstitially. Oxide minerals are rare. In general, the abundance of biotite decreases as arfvedsonite content increases. Flow lineation is formed by alignment of lath-shaped arfvedsonite. Banded samples contain zones of anhedral, poikilitic arfvedsonite intergrown with zircon and fluorite.

Late-stage transitional ferrobasalts

These dikes contain phenocrysts of olivine and titanaugite in diabasic to intergranular groundmass of plagioclase laths ($An_{50–60}$), olivine, titanaugite, ilmenite, and apatite. Some dikes contain interstitial alkali feldspar. This assemblage is locally recrystallized to a granular aggregate of olive green amphibole, plagioclase, and oxides.

MINERAL COMPOSITIONS

Feldspars

Measured compositions (Table 2) from unexsolved alkali feldspar in quartz syenites were within a few mole percent of $An_5Ab_{59}Or_{36}$. Calculated alkali feldspar compositions (see Appendix A for method) range from $An_4Ab_{58}Or_{38}$ in quartz syenite to $An_1Ab_{54}Or_{45}$ in stage 2 granite (Shannon, 1994). Plagioclase cores in alkali feldspar crystals are as calcic as $An_{36}Ab_{62}Or_2$ in stage 2 granite and range from $An_{24}Ab_{75}Or_1$ to $An_{32}Ab_{66}Or_2$ in stage 3 quartz syenite.

Pyroxene

Pyroxene is ferrohedenbergite with the compositional range of $Di_{42}Fs_{49}En_8$ to $Di_{46}Fs_{52}En_3$ (Table 3). Mn content ranges from 0·01 to 0·03 per formula unit (p.f.u.).

Amphibole

Calculated Fe^{3+}/Fe^{2+} ratios (Spear & Kimble, 1984) range from 0·00 to 0·19 for stages 1–4, and from 0·21 to 0·75 for stage 5 (Table 3). Magmatic amphibole in stages 1–4 ranges from ferroedenitic hornblende through ferroedenite to ferroactinolite (Fig. 4). Amphibole in stage 5 is arfvedsonite.

Calcic amphibole compositions were controlled by a combination of edenite (Fig. 4) and Ti-Tschermaks substitutions. Riebeckite and richterite substitution was not apparent. If the hydroxyl site is filled, the ratio (F + Cl)/(OH + F + Cl) in ferroedenite is ~0·25. The fluorine content of the ferroedenitic amphiboles decreases with increasing Fe/(Fe + Mg), where Fe indicates total iron content.

Arfvedsonite from stage 5 (and metasomatized stage 1 sills) is low in Al: 0·07–0·25 cations p.f.u. (Table 3). As Fe/(Fe + Mg) increases from 0·84 to 1·00, Ti decreases from 0·25 to 0·05 cations p.f.u. The ratio F/(OH + F + Cl) for arfvedsonite ranges from 0·35 to 0·87.

In general, ferroedenitic amphibole has higher F/(OH + F + Cl) than coexisting biotite (Tables 3 and 4).

Table 2: Representative feldspar compositions

	Oxide composition							
Analysis:	1	2	3	4	5	6	7	8
Mineral:	alk feld	alk feld	alk feld	alk feld	plag	plag	alk feld	albite
Sample:	144-1	153-5	519-5	132-1	153-8c	153-8r	519-3c	519-3r
Stage:	1	3	3	4	3	3	3	3
SiO_2	65·19	66·07	65·73	65·13	61·46	62·09	66·32	68·00
Al_2O_3	22·02	19·50	19·28	18·12	23·95	24·06	19·57	19·84
FeO	0·25	0·07	0·07	0·12	0·05	0·07	0·04	0·19
CaO	1·29	0·94	0·71	0·00	5·72	5·37	0·79	0·70
Na_2O	6·46	6·67	6·34	0·68	8·23	8·52	8·09	10·73
K_2O	6·59	6·26	6·56	15·66	0·15	0·17	4·04	0·34
BaO	0·30	1·39	1·26	0·07	0·00	0·41	0·56	0·00
Total	100·10	100·90	99·95	99·78	99·56	100·69	99·41	99·80
	Cations per 8 oxygens							
Si	2·931	2·958	2·968	3·011	2·739	2·744	2·968	2·979
Al	1·061	1·029	1·026	0·987	1·258	1·253	1·032	1·025
Fe	0·008	0·002	0·002	0·004	0·002	0·002	0·001	0·006
Ca	0·062	0·045	0·034	0·000	0·273	0·254	0·038	0·033
Na	0·563	0·579	0·555	0·061	0·711	0·730	0·702	0·912
K	0·378	0·358	0·378	0·924	0·009	0·010	0·231	0·019
Ba	0·005	0·024	0·022	0·001	0·000	0·007	0·010	0·000
Total	5·009	4·996	4·986	4·988	4·992	5·000	4·982	4·974

Analyses 1, 2, 3, and 7 are area-averaged analyses from feldspars with fine exsolution lamellae. Analyses 4 and 8 are of exsolved microcline and albite, respectively. Analyses 5 and 6 are from unexsolved plagioclase.

This relationship seems unusual in that F is typically considered to be enriched in the melt during crystallization. However, petrographic relationships show that amphibole crystallized before biotite and that magmatic fluorite coprecipitated with biotite. Fluorite stability apparently had the effect of greatly reducing the fluorine available to coprecipitating biotite.

Micas

The biotite is annite; Fe/(Fe + Mg) ranges from 0·84 to 1·00 (Fe as total iron). The ratio increases from stage 3 quartz syenite to arfvedsonite granite. Mn content is low and nearly constant; Ti decreases from 0·45 to 0·15 cations p.f.u. as Fe/(Fe + Mg) increases (Table 4). The Fe/(Fe + Mg) of coexisting annite and amphibole are nearly identical to each other and to whole-rock Fe/(Fe + Mg), an indication that whole-rock Fe/(Fe + Mg) is controlled by mafic silicates. Chlorine typically fills ~4% of the hydroxyl site; fluorine ranges from zero to 7% site occupancy. However, in annite from stage 5 arfvedsonite granite, F/(F + Cl + OH) is 0·35.

Tetrahedral Si in biotite ranges from 5·59 cations p.f.u. in stage 3 to 6·57 in stage 5. Only in stage 5 does annite contain excess Si^{IV}. This excess may be explained by exchange on the tetrahedral site of the form: $Si^{IV} + (Fe^{2+}, Mg)^{VI} = 2Al^{IV} + \square^{VI}$, which is common in peralkaline magmas (Czamanske & Dillet, 1988). In the remaining stages (where Si^{IV} in biotite is <6·00), Al^{IV} is positively correlated with Ti^{VI}. This correlation is probably a function of Ti-Tschermaks substitution and substitution of Ti^{VI} for octahedral vacancies (Guidotti, 1984; Abrecht & Hewitt, 1988).

White mica in the topaz-bearing stage 4 leucogranite has Fe/(Fe + Mg) >0·97 and Mn/(Mn + Ti) as high as 0·45 (not shown). The F/(F + Cl + OH) value is 0·55. If this mica's formula is calculated such that all Fe is Fe^{3+}, it would be classified as ferrimuscovite. However, the whole-rock Fe^{3+} content is close to zero (see below), so that the mica is probably not ferrimuscovite but

Table 3: Representative compositions of amphibole and clinopyroxene

	Oxide composition									
Mineral:	amph	amph	amph	amph	amph	amph	amph	amph	cpx	cpx
Sample:	144-8	549-8	602-2a	153-2	519-2	132-3	542-1	543-6	144-1a	916-2
Stage:	1	2	2	3	3	4	5	5	1	2
SiO_2	41·35	39·83	41·05	42·62	40·23	42·52	50·05	50·23	47·57	48·95
TiO_2	2·01	2·02	1·28	1·82	2·36	1·52	0·98	0·74	0·88	0·07
Al_2O_3	6·99	7·68	6·53	6·11	7·93	5·71	0·72	0·58	0·95	0·24
FeO	31·30	32·64	32·66	32·14	31·05	33·73	33·32	31·73	27·75	26·68
MnO	0·51	0·43	0·55	0·65	0·81	0·60	0·53	0·52	0·77	0·91
MgO	1·36	1·36	0·95	1·73	1·98	1·02	0·03	0·36	1·81	1·20
CaO	9·16	10·09	9·86	9·70	10·39	9·27	0·11	1·00	18·50	18·61
Na_2O	2·09	2·21	2·28	2·20	2·24	2·36	8·99	8·44	0·42	2·54
K_2O	1·22	1·33	1·18	1·04	1·29	1·06	1·09	1·48	nd	0·00
Cl	0·70	0·82	0·39	0·34	0·41	0·28	0·02	0·00	nd	nd
F	1·16	1·18	0·37	0·24	0·88	0·41	2·51	2·66	nd	nd
Total	98·28	99·69	98·56	98·76	99·67	98·62	99·19	98·83	98·77	99·86
	Cations per 23 oxygens							*Cations per 6 oxygens*		
Si	6·75	6·54	6·66	6·87	6·51	6·93	8·00	8·00	1·96	1·98
Ti	0·25	0·25	0·16	0·22	0·29	0·19	0·12	0·09	0·03	0·00
Al	1·34	1·48	1·44	1·16	1·51	1·10	0·14	0·11	0·05	0·01
Fe^{3+}	0·53	0·13	0·56	0·20	0·12	0·17	1·01	1·31	0·03	0·20
Fe^{2+}	3·74	4·27	3·87	4·13	4·08	4·37	3·44	2·92	0·93	0·70
Mn	0·07	0·06	0·08	0·09	0·11	0·08	0·07	0·07	0·03	0·03
Mg	0·33	0·33	0·23	0·42	0·48	0·25	0·01	0·09	0·11	0·07
Ca	1·60	1·77	1·71	1·68	1·80	1·62	0·02	0·17	0·82	0·81
Na	0·66	0·70	0·72	0·69	0·70	0·75	2·78	2·13	0·03	0·19
K	0·25	0·28	0·24	0·21	0·27	0·22	0·22	0·30	nd	nd
Cl	0·19	0·23	0·11	0·09	0·11	0·08	0·01	0·00	nd	nd
F	0·60	0·61	0·19	0·12	0·45	0·21	1·27	1·34	nd	nd
Total	16·31	16·65	15·97	15·88	16·43	15·97	17·09	16·53	3·99	3·99

nd, not determined. Fe^{3+} and Fe^{2+} calculated according to Spear & Kimball (1984). Sites were assigned according to Leake (1978) and Hawthorne (1983).

zinnwaldite, which is common in topaz–fluorite–cassiterite-bearing pegmatites (Deer *et al.*, 1966). If so, the calculated octahedral vacancies total 1·06, some or all of which could be filled by Li.

Astrophyllite

Astrophyllite analyses (Table 5) yield the average formula of $(Ca_{0·06}Na_{1·20}K_{1·38})(Mg_{0·02}Mn_{0·33}Fe_{6·50})^{VI}Ti_{1·71}{}^{VI}(Al_{0·12}Si_{7·85})$ $^{IV}O_{24}[(O,OH)_{6·1},F_{0·85},Cl_{0·01}]$. Octahedral Ti is <2 cations p.f.u., which suggests that an unanalyzed element such as Zr or Nb is present in that site (e.g. Platt & Woolley, 1986).

Zircon

Zircons (Table 6) have 0·85–2·5% HfO_2 with average Zr/Hf ratios of $37·2 \pm 9·7$. Zr/Hf tends to be lower in zircons from stages 2 and 4; however, as much compositional variability exists within zircons from a single sample as exists between the intrusive stages.

Cores and oscillatory zones in zircon mantles are enriched in Y, Ca, Al, and Fe. These zones contain up to 3·5% Y_2O_3, 0·5% CaO, 0·5% Al_2O_3 and 1·0% FeO, they are optically murky ('metamict'), and yield low analytical totals (90–98 wt %). Clear mantle zones and rims are stoichiometric, give 100% totals, have the highest Zr/Hf ratios, and the lowest Hf, Y, Ca, Al, and Fe contents. The presence of 'metamict' zones in zircons can explain the observed Pb loss.

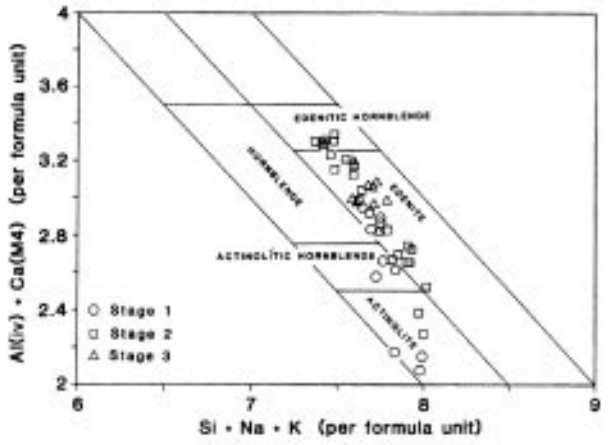


Fig. 4. Composition and classification of calcic amphiboles in stages 1–3 (Giret *et al.*, 1980).

Caruba *et al.* (1985) showed that, in the presence of fluorine, hydroxylated zircon can form at magmatic temperatures, and they concluded that hydroxylated zircon is more susceptible to radiation damage. It is also possible that hydroxylated zircon is able to accept higher concentrations of Y, Ca, Al, and Fe. The oscillatory nature of 'metamict' zones suggests that this feature developed at magmatic temperatures, possibly by hydroxylation in the presence of F-rich magma. If so, oscillatory zoning implies variations in the activity of fluorine, or in the diffusion of a rate-limiting constituent. The clear, stoichiometric rims suggest that the activity of fluorine was lowered, probably by crystallization of fluorite, before zircon crystallization ceased.

CONDITIONS OF EMPLACEMENT

Pressure

The minimum pressure of emplacement was obtained from stratigraphic reconstruction. The stratigraphic thickness of 1175–1600 m of the Proterozoic sedimentary and volcanic section, weighted by appropriate densities for each rock type, corresponds to pressures of ~0·31–0·43 kbar. It is noteworthy that the subsolvus character of stages 2 and 3 implies a stage of crystallization at pressures greater than the level of emplacement and at water-undersaturated conditions (Nekvasil, 1990).

Volatile contents

Water

The paucity of pegmatitic rocks in stages 1–4 suggests that the magmas were near the solidus before fluid saturation occurred. An estimate of the maximum water content of the least differentiated granitic magmas can be made if one assumes that the magmas were related by fractional crystallization at ~0·4 kbar pressure (see below) and that the most evolved leucogranitic magmas reached H_2O saturation. At 0·5 kbar, a water-saturated granitic melt of ternary minimum composition contains 2·3–2·9 wt % H_2O (Tuttle & Bowen, 1958). Mass balance calculations (see below) show that stage 4 leucogranitic magmas could be derived from stage 3 quartz syenitic magmas by 85–90% crystallization. This type of mass balance calculation indicates that a parental stage 3 syenitic magma contained no more than 0·3 wt % water. Thus the syenitic magmas were probably undersaturated, because saturation in the alkali feldspar–H_2O system at 500 bars requires ~2 wt % H_2O at temperatures below ~930°C (Tuttle & Bowen, 1958).

Fluorine

Petrographic evidence for fluorite as a late magmatic phase in all stages, and the moderate fluorine content in amphibole, zinnwaldite, astrophyllite, and stage 5 biotite all attest to the presence of fluorine in the magmas. In a plot of normative Ab–Qtz–Or (Fig. 5), evolved granitic compositions (stage 1 granite, stages 4 and 5) converge near a low-pressure water-saturated minimum, at which point the stage 4 and 5 compositional trends turn toward sodium enrichment. Such Na enrichment was documented for fluorine-bearing H_2O-saturated granitic melts at 1 kbar (Manning, 1981; Pichavant & Manning, 1984).

An estimate of fluorine fugacity relative to Cl and H_2O can be made on the basis of Munoz' (1984) activity models for biotite. For biotite in stages 2–5, these models indicate that $f(HF)/f(H_2O)$ ranged from $10^{-5\cdot2}$ to $10^{2\cdot5}$, $f(HCl)/f(H_2O)$ ranged from $10^{-2\cdot7}$ to $10^{-1\cdot3}$, and $f(HF)/f(HCl)$ ranged from $10^{-3\cdot5}$ to $10^{+0\cdot1}$. These values are typical of fluorite-buffered magmas at low $f(O_2)$ (Munoz, 1969, 1984).

Temperature

Zircon saturation temperature

Temperature was estimated by applying the zircon saturation thermometer of Watson & Harrison (1983). This method is thought to be appropriate because euhedral zircon is present in all stages. In addition, no Pb isotopic evidence for inherited zircon exists. Quartz syenitic dikes of stage 3, some of the least differentiated samples of the RBGS, and trachytic to rhyolitic rocks of the comagmatic Thunderbird Group contain euhedral zircon microphenocrysts, which suggests that zircon was in equilibrium with RBGS melts early in their history. Therefore, estimated temperatures are thought to reflect near-liquidus conditions.

Table 4: Representative compositions of biotite

	Oxide composition				
Sample:	629-9	157-2	519-3	132-5	623-2
Stage:	1	2	3	4	5
SiO_2	35·39	34·30	35·21	34·89	39·19
TiO_2	2·38	1·21	3·48	2·76	1·85
Al_2O_3	10·34	10·32	11·55	10·77	7·85
FeO	38·57	40·11	34·94	37·26	35·60
MnO	0·47	0·35	0·40	0·33	0·32
MgO	0·50	0·42	1·95	0·33	0·09
CaO	0·00	0·05	0·02	0·11	0·00
Na_2O	0·03	0·03	0·03	0·00	0·00
K_2O	7·65	8·16	8·30	8·12	8·25
Cl	0·78	0·78	0·48	0·39	0·02
F	0·03	0·00	0·04	0·00	3·47
Total	96·14	95·73	96·4	94·96	96·64
	Cations per 22 oxygens				
Si	5·86	5·78	5·71	5·81	6·52
Ti	0·30	0·15	0·42	0·35	0·23
Al	2·02	2·05	2·21	2·11	1·54
Fe^{3+}	0·64	0·67	0·57	0·63	0·59
Fe^{2+}	4·70	4·98	4·17	4·57	4·36
Mn	0·07	0·05	0·05	0·05	0·05
Mg	0·12	0·11	0·47	0·08	0·02
Ca	0·00	0·01	0·00	0·02	0·00
Na	0·01	0·01	0·01	0·00	0·00
K	1·62	1·76	1·72	1·73	1·75
Cl	0·22	0·22	0·13	0·11	0·01
F	0·02	0·00	0·02	0·00	1·83
Total	15·58	15·79	15·48	15·46	16·90

Fe^{3+} and Fe^{2+} calculated according to the ratio $Fe^{3+}/Fe^{2+} = 0{\cdot}136$ (e.g. Guidotti & Dyar, 1991); oxyannite component calculated to balance Fe^{3+}, and OH^- calculated by difference.

Calculated temperatures for the granitic rocks range from ~1060°C to 730°C and decrease as a function of increasing SiO_2. Temperatures from ~1060°C to 870°C were obtained from samples of the Thunderbird Group.

Apatite saturation thermometry

Apatite saturation temperatures (Harrison & Watson, 1984) are as much as 200°C lower than zircon saturation temperatures. They do not correlate with any differentiation index. These results suggest that apatite became saturated below the liquidus and are consistent with the apatite's interstitial habit.

Compositional trend in the haplogranite system

The trend of RBGS compositions in a plot of normative albite, quartz, and orthoclase (Fig. 5) is parallel to the thermal trough determined by Tuttle & Bowen (1958). At 500 bars pressure and H_2O-saturated conditions, this trend corresponds to maximum temperatures of ~925°C at the Ab–Or side line and ~780°C at the ternary minimum. However, petrographic and mass balance evidence for H_2O-undersaturated conditions suggest temperatures ≥ 1000°C for syenitic compositions near the Ab–Or sideline (Tuttle & Bowen, 1958). These temperatures are consistent with those estimated from zircon saturation thermometry.

Table 5: Representative compositions of astrophyllite

	Oxide composition		
Sample:	543-6	543B-10	543B-6
SiO_2	35·82	35·52	35·94
TiO_2	10·77	10·20	10·84
Al_2O_3	0·38	0·49	0·29
FeO	35·47	35·85	35·63
MnO	1·70	1·77	1·82
MgO	0·05	0·05	0·07
CaO	0·29	0·29	0·39
Na_2O	2·91	2·87	4·98
K_2O	4·88	4·65	4·98
Cl	0·02	0·02	0·02
F	1·51	1·27	1·45
Total	93·80	92·98	94·32
	Cations per 27·5 oxygens		
Si	7·82	7·82	7·81
Ti	1·77	1·69	1·77
Al	0·10	0·13	0·07
Fe	6·48	6·60	6·48
Mn	0·31	0·33	0·34
Mg	0·02	0·02	0·02
Ca	0·07	0·07	0·09
Na	1·23	1·23	1·22
K	1·36	1·31	1·38
Cl	0·01	0·01	0·01
F	1·04	0·88	1·00
Total	20·21	20·09	20·19

In summary, we interpret the temperatures of emplacement of syenitic and granitic rocks of the RBGS to range as high as 1060°C. Such high temperatures are borne out by the results of the zircon saturation thermometer on both granitic and crystal-poor volcanic rocks, the compositional trend of syenitic rocks in the haplogranite system, and the high temperatures of contact metamorphism at such low pressure.

Oxygen fugacity

The iron-rich nature of the mafic silicates, the paucity of magnetite in many samples, and the absence of titanite indicate crystallization from low $f(O_2)$ magma (e.g. Wones, 1981, 1989; Bonin & Giret, 1985). An upper $f(O_2)$ limit is defined by the assemblage: hedenbergite, ilmenite, quartz, $\pm$magnetite (without titanite), which is stable at $f(O_2)$ no higher than two log units above the fayalite–magnetite–quartz buffer (FMQ; Wones, 1989; Fig. 6). If fayalite was a primary mineral in the suite, as indicated by late-stage grunerite, then $f(O_2)$ conditions were near FMQ. In addition, Fe-rich grunerite stability at low T requires $f(O_2)$ conditions below the Ni–NiO buffer (e.g. Gilbert *et al.*, 1982; Fig. 6). Finally, arfvedsonite stability is limited to $f(O_2)$ near or below FMQ (Ernst, 1962). Thus, as shown in Fig. 6, the T–$f(O_2)$ path for the RBGS was close to FMQ throughout crystallization.

GEOCHEMISTRY

Granitic and syenitic rocks

Major and trace element compositions (Table 7) vary continuously from quartz syenitic to leucogranitic compositions. Over a range of silica contents (in wt %) from 61 to 78% (Fig. 7), total alkali content decreases from 11 to 7·5%, such that the RBGS trend crosses the alkaline–subalkaline boundary (Irvine & Baragar, 1971) at ~68% SiO_2 (Fig. 7). For the suite, CaO and MgO

Table 6: Representative compositions of zircon

	Oxide composition						
Sample:	144-2c*	157-6t*	157-6c	153-2c	153-3t	542-1c	542-1t
Stage:	1	2	2	3	3	5	5
SiO_2	31·39	31·03	31·91	31·71	31·46	31·51	30·47
Al_2O_3	0·01	0·13	0·00	0·00	0·00	0·00	0·02
FeO	0·09	0·57	0·00	0·13	0·06	0·11	0·71
CaO	0·02	0·61	0·05	0·02	0·04	0·01	0·20
ZrO_2	66·72	60·94	66·54	65·92	64·07	66·07	62·17
HfO_2	1·44	1·91	1·94	1·25	1·41	2·42	1·45
Y_2O_3	0·04	0·96	0·09	0·18	1·16	0·00	1·04
Total	99·72	96·21	100·53	99·21	98·20	100·12	96·06
	Cations per 4 oxygens						
Si	0·974	0·993	0·982	0·985	0·989	0·977	0·980
Al	0·000	0·005	0·000	0·000	0·000	0·000	0·001
Fe	0·002	0·002	0·000	0·003	0·002	0·003	0·019
Ca	0·001	0·021	0·002	0·001	0·001	0·000	0·007
Zr	1·010	0·951	0·999	0·999	0·982	0·999	0·975
Hf	0·013	0·017	0·017	0·011	0·013	0·021	0·013
Y	0·001	0·016	0·001	0·003	0·019	0·000	0·018
Total	2·001	2·005	2·001	2·002	2·006	2·000	2·013

*c indicates clear zones in crystals; t indicates turbid zones.
Y, Ca, Al, and Fe assigned to the Zr site.

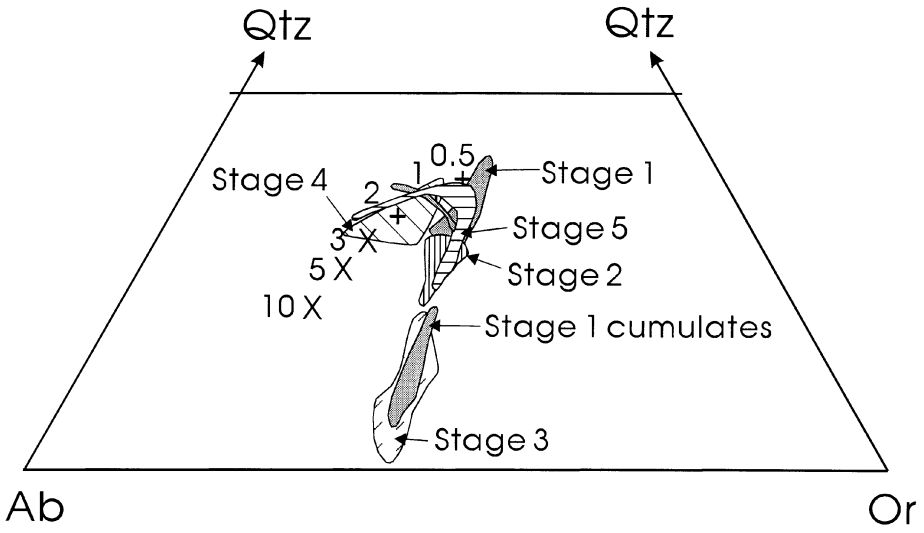


Fig. 5. Compositional ranges of the RBGS in the normative Ab–Or–Qtz ternary. Stage 1 cumulates are alkali feldspar cumulates from the centers of the stage 1 sills in Fusselman Canyon. +, H_2O-saturated minima at 0·5, 1, and 2 kbar (Tuttle & Bowen, 1958); ×, H_2O-saturated eutectics at 3, 5, and 10 kbar (Luth *et al.*, 1964).

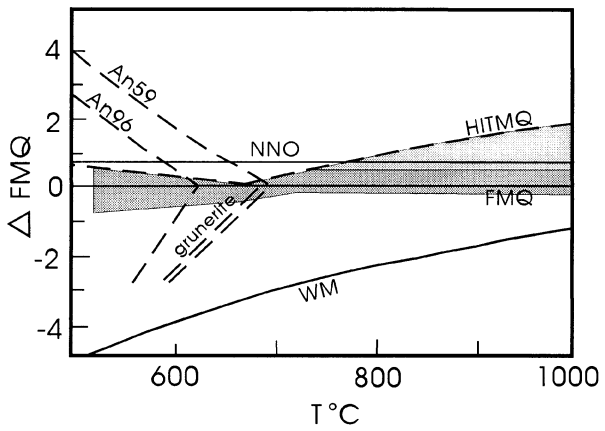


Fig. 6. Estimate of oxygen fugacity in terms of ΔFMQ, the log difference in $f(O_2)$ relative to FMQ (the fayalite–magnetite–quartz buffer), plotted as a function of T, at P_{total} = 430 bars. Buffer curves shown for Ni–NiO (NNO), wüstite–magnetite (WM), and hedenbergite–ilmenite–titanite–magnetite–quartz (HITMQ). Maximum T stability for grunerite shown by short-dashed lines; maximum T stability for annitic biotite with limiting activities of 0·96 and 0·59 shown by long-dashed lines (Wones, 1981). Inferred T–$f(O_2)$ path of the RBGS indicated by shading; the denser pattern shows the most likely path: near FMQ.

contents are low (<3% and <0·7%, respectively; Table 7). In stages 1–4, as SiO_2 increases, Al_2O_3 decreases from 17 to 11%, total Fe (as Fe_2O_3) decreases from 10 to 0·5% (Fig. 8), and Mg/(Mg + Fe) decreases from 0·1 to 0·001. These features (high total Fe, low CaO and MgO, and low *mg*-number) are typical of A-type granites (Collins *et al.*, 1982; Whalen *et al.*, 1987; Anderson & Bender, 1989; Eby, 1992).

Most of the suite is metaluminous; although a few samples from stages 1 and 4 are peraluminous. Most peraluminous samples are fine-grained dikes or sill margins and have probably lost alkalis. A few samples of stage 4 are peralkaline, with (Na + K)/Al values slightly greater than one. However, within analytical uncertainty, all stage 4 samples can be considered to be metaluminous.

In contrast, all but one sample of stage 5 are peralkaline, with (Na + K)/Al values as high as 1·38. As peralkalinity increases in this group, SiO_2 decreases and total Fe increases (e.g. Bonin, 1986). Therefore, for stages 1–4, SiO_2 content is used as a differentiation index but for stage 5 the (Na + K)/Al value is used.

Fe^{3+}/Fe_{Total} values form broad clusters for stages 1, 2, 3, and 5. Because subsolidus oxidation is common, the lowest Fe^{3+}/Fe_{Total} values are taken to indicate the magmatic value. These values are ~0·1 for stage 1, ~0·2 for stages 2, 3, and 4, and ~0·4 for stage 5. These values are consistent with the low $f(O_2)$ determined from mineral equilibria.

Relative to calc-alkaline granites, the RBGS shows high concentrations of high field strength elements (HFSE) and large ion lithophile elements (LILE). From quartz syenite to alkali feldspar granite, as SiO_2 increases, Cs, Rb, Th, U, Be, and the light rare earth elements (LREE) increase; Ta, Nb, Y, and the heavy rare earth elements (HREE) remain approximately constant; and Zr, Hf, Ti, P, Sc, V, Zn, Mn, Ba, and Sr decrease (Figs 9–13; Table 7). Most of the minor and trace elements have a relatively large (more than four-fold) range of variation from quartz syenitic to granitic compositions (e.g. Figs 9 and 10). Although the REE and Y have a smaller range of variation among metaluminous samples (less than three-fold), they show more than six-fold variation among peralkaline samples (e.g. Fig. 11e).

Stage 1 compositions can be distinguished from those of stages 2–4 in several plots. Relative to the trend defined by samples from stages 2, 3, and 4, the trend defined by stage 1 samples is lower in Al_2O_3 (Fig. 8) and higher in CaO (not shown). In addition, stage 1 quartz syenite is generally lower in total alkalis (Fig. 7) and higher in TiO_2 than stage 3 quartz syenite. This relationship suggests that stages 2–4 may be comagmatic but that stage 1 is probably not.

Chondrite-normalized REE patterns are characterized by shallow negative slopes and negative Eu anomalies (Fig. 11). With increasing differentiation from stage 3 to stage 2 (increase in SiO_2), overall REE abundance increases and the Eu anomaly deepens. The REE in stage 4 samples decrease with increasing SiO_2 and Rb, and their patterns display a distinctive depletion of middle REE (MREE) (Fig. 11). In contrast, samples from stage 5 are progressively enriched in REE (Fig. 11), and particularly the MREE, with differentiation [as defined by increasing peralkalinity: (Na + K)/Al]. The size of the negative Eu anomaly (Eu/Eu*) is well correlated with Ba and Sr among stages 1, 2, and 3 (Fig. 12); all values decrease with increasing differentiation. However, Eu/Eu* is constant among samples from stage 4 (Fig. 12).

Mantle-normalized multielement diagrams (Fig. 13) emphasize the similarity of elemental ratios within each intrusive stage. They also demonstrate the progressive decrease of P_2O_5, Ba, Sr, Eu, and TiO_2 with increasing differentiation.

Isotopic data for the RBGS are presented elsewhere and summarized by Smith *et al.* (1997). Neodymium, lead, and oxygen isotopic compositions vary within narrow ranges that are consistent with a comagmatic relationship. Limited Nd isotopic data are available for late-stage ferrobasalts, and they are identical to those for the granitic rocks.

Transitional ferrobasalts

These dikes are treated as a group and are assumed to be Proterozoic in age, although only two show field

Table 7: Representative major and trace element analyses

Sample:	TM2A	107	170	711	549	133	602	151-3	538	532	714	519	726	136	551	146	542	MK1	623	9401
Stage:	1	1	1	2	2	2	2	2	3	3	3	3	4	4	4	4	5	5	5	6
wt %																				
SiO_2	63·97	74·73	74·86	68·9	71·77	73·31	73·99	74·9	62·89	63·71	64·52	64·66	76·07	76·62	76·92	77·65	71·01	74·14	76·95	46·4
TiO_2	0·63	0·23	0·24	0·37	0·36	0·2	0·21	0·15	0·54	0·56	0·9	0·58	0·15	0·12	0·02	0·07	0·21	0·14	0·1	3·84
Al_2O_3	14·54	11·98	11·94	14·32	13·09	13·07	12·55	12·24	16·19	15·66	14·49	15·08	12·18	11·84	12·98	11·93	8·94	9·34	11·46	15·88
Fe_2O_3	1·45	0·81	0·58	1·85	0·89	1·41	0·92	0·78	1·75	1·7	2·85	1·47	1·05	1·41	−0·04	0·61	4·23	2·59	1·02	14·76
FeO	5·57	2·34	2·55	2·67	2·1	0·95	1·77	1·38	4·41	4·49	4·54	4·44	0	0·36	0·57	0·39	5·24	3·24	1·44	nd
MnO	0·17	0·07	0·05	0·1	0·05	0·04	0·06	0·05	0·19	0·18	0·14	0·17	0·04	0	0·01	0·01	0·13	0·09	0·03	0·21
MgO	0·23	0·16	0·18	0·08	0·12	0·04	0·06	0·03	0·31	0·4	0·69	0·41	0·02	0·04	0·02	0·04	0·07	0·05	0·03	4·6
CaO	2·2	0·7	0·76	1·19	1·02	0·69	0·85	0·9	2·28	1·91	2·26	2·03	0·43	0·46	0·16	0·53	0·58	0·6	0·17	7·16
Na_2O	4·72	3·16	3·74	3·92	3·69	4·31	3·82	3·74	5·49	4·99	3·18	5·03	3·2	3·9	5·44	3·73	4·74	4·01	4	3·65
K_2O	5·5	5·06	4·25	6·24	5·83	5·57	5·36	4·86	5·41	5·62	5·57	5·52	5·8	4·67	3·56	4·57	4·21	4·33	4·32	1·39
P_2O_5	0·09	0·03	0·02	0·04	0·05	0·02	0·03	0·03	0·09	0·11	0·26	0·12	0·02	0·01	0·02	0·02	0·01	0·01	0·01	0·72
LOI	0·34	0·99	1·04	0·55	0·41	0·71	0·45	0·41	0·44	0·41	1·1	0·39	0·56	0·63	0·23	0·6	0·16	0·26	0·35	
Total	99·4	100·24	100·21	100·22	99·37	100·32	100·06	99·46	99·99	99·73	100·47	99·9	99·53	100·07	99·88	100·15	99·53	98·8	99·88	98·61
p.p.m.																				
Cs	2·6	0·4	0·9	3·2	4	2·9	3·2	2·2	1·8	1·4	2·1	1·6	0·4	0·9	3·4	4·2	2·4	2·8	2·2	bd
Rb	153	166	114	260	257	303	282	300	139	175	220	196	231	293	840	638	649	707	444	168
Sr	82	38	25	33	43	22	29	32	121	80	139	68	13	7	9	15	33	22	8	262
Ba	932	268	262	581	486	176	290	266	1534	1164	1101	978	167	20	37	70	16	24	18	294
Be	nd	nd	nd	nd	nd	nd	nd	5·41	6·06	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd
Zr	1210	661	617	949	677	761	539	346	1748	1345	941	1287	519	235	101	190	1018	917	537	221
Hf	29·8	20·3	20·2	25·8	19·5	24·3	17·7	13·8	41·8	33·4	22·4	31·7	18·8	10·4	15	11·2	44·9	36·8	22·2	5·8
Y	103	154	166	139	125	203	137	145	121	130	128	141	219	192	33	114	1307	405	260	60·4
Nb	59	74	75	66	50	99	67	43	86	93	64	94	94	74	110	107	165	258	147	5
Ta	2·8	4·1	3·9	3·6	3·1	6·3	4·5	2·7	3·6	4·1	3·1	4·8	5·3	3·7	18·3	11·5	9·2	12·3	8·3	0·71
Sc	7·4	1·3	1·3	4·3	4·2	2·2	2·2	1·7	15·5	14·9	10·1	15·7	1·5	0·1	0·5	1·1	0·4	0·5	0·2	34·8
V	5	2	2	2	4	2	2	5	3	3	23	3	1	1	1	1	2	3	4	242
Cr	6	bd	bd	1	2	2	1	3	14	bd	3	4	bd	bd	1	bd	2	2	8	101
Ni	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	142
Cu	9	3	5	5	12	7	4	13	5	7	2	8	7	3	2	7	4	6	2	40
Zn	160	190	130	154	157	125	92	52	182	188	80	192	37	94	23	19	1186	856	381	140
U	2·8	6·7	7·3	3·9	4·8	7·2	5·3	8·6	4·1	3·6	4	4·1	7·3	3·1	6·1	9·9	9·7	11·4	6·8	bd
Th	11·3	30·4	30·1	24	27·2	35·3	28·4	36·8	14·9	17·1	19·1	16·7	28	8·1	13·5	52·8	22·2	37·2	23·3	3·4
La	87·6	133	133·4	94·8	147	217	116	104	76·6	102	92·3	128	140	70·4	22·4	60·8	527	199	74·3	21·8
Ce	220·2	327	336·8	221	271	462	267	255	186	263	216·2	291	326	154	60·9	150	1422	568	170	48·5
Nd	113	148	153·9	117	138	193	117	98·2	106	128	109·7	149	145	65	19·5	42·7	679	260	70·4	31·2
Sm	22·9	29·6	30·6	25·4	28·4	37·3	23·7	20·8	23·2	27·3	23·7	30·5	32·3	14·2	4·13	10·6	205	68·9	19·4	8·2
Eu	3·83	1·64	1·54	2·06	2·12	1·87	1·48	1·02	5·03	4·34	3·79	4·29	0·94	0·47	0·14	0·36	4·16	1·48	0·67	2·29
Tb	2·92	4·88	4·64	3·64	3·57	5·66	3·35	3·43	4·05	4·06	3·41	4·58	4·69	3·16	0·73	1·8	43·9	14·9	5·24	1·5
Yb	13	17·6	17·01	16	12·6	20·7	15·8	16·8	14·1	15·8	14·9	16·7	21·7	11·1	7·11	18·3	52·2	34·9	21·8	4·62
Lu	2·01	2·46	2·31	2·49	1·87	2·86	2·27	2·28	2·23	2·47	2·3	2·45	3·22	1·39	1·13	2·74	7·37	5·28	2·98	0·76

nd, not determined; bd, below detection limits.

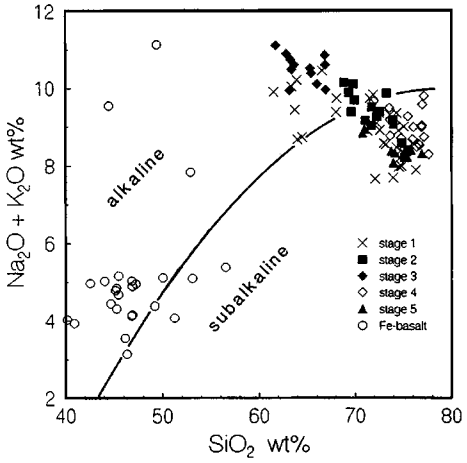


Fig. 7. Variation of total alkalis in the RBGS and ferrobasaltic dikes.

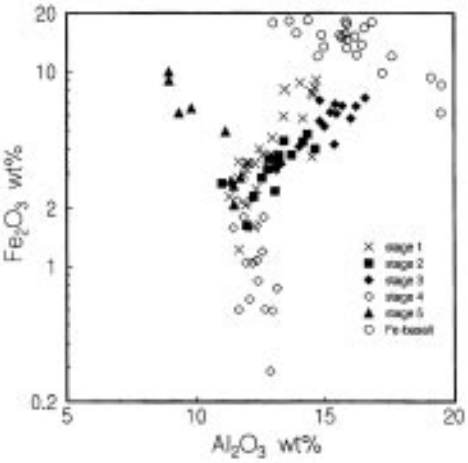


Fig. 8. Variation of Al_2O_3 as a function of total Fe expressed as Fe_2O_3. (Note that stage 1 samples plot on a distinct trend relative to stages 2–4.)

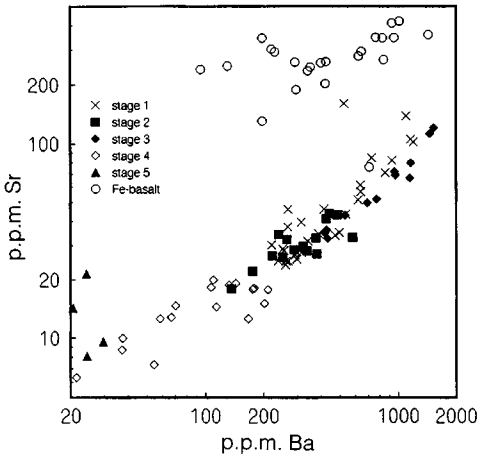


Fig. 9. Variation of Sr as a function of Ba content.

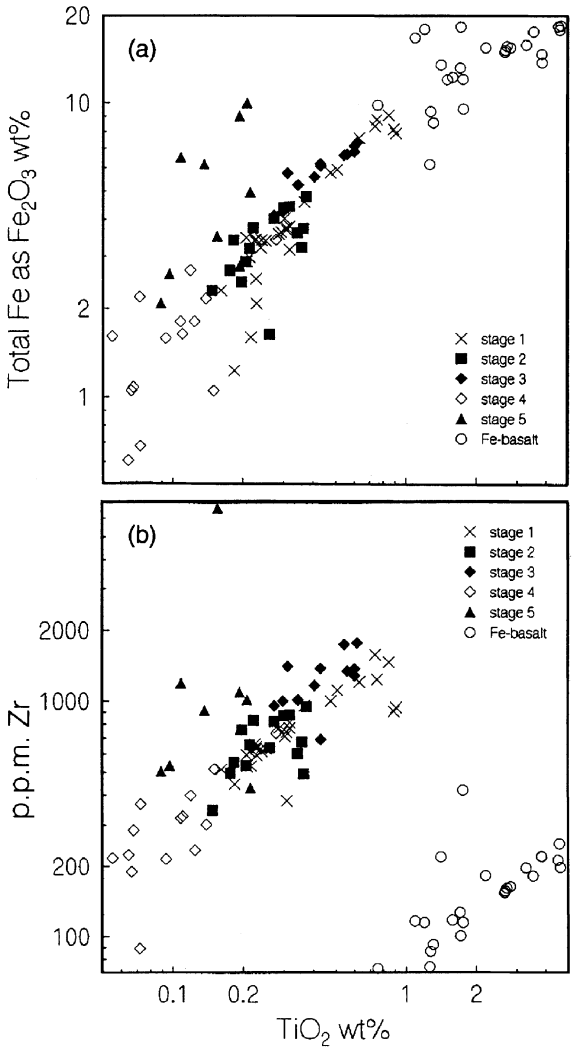


Fig. 10. Variation of Fe_2O_3 (a) and Zr (b) as a function of TiO_2.

evidence for this age. However, their overall compositional similarities support this assertion. These compositions are Fe rich, plot near the alkaline–subalkaline boundary (Fig. 7; Irvine & Baragar, 1971), and range from basalt to trachybasalt. The use of alkali contents to classify these dikes is suspect because of possible post-emplacement mobility of alkalis and alkaline-earths (note scatter in Fig. 7). Samples are LREE enriched (Fig. 11); however, normalized La/Lu values are in the range 3–6; less than is typical of alkaline basalts (Thompson *et al.*, 1984; Sun & McDonough, 1989). Therefore, we refer to these dikes as transitional ferrobasalts.

Mantle-normalized multielement patterns (not shown) are generally similar to ocean island basalt (OIB)

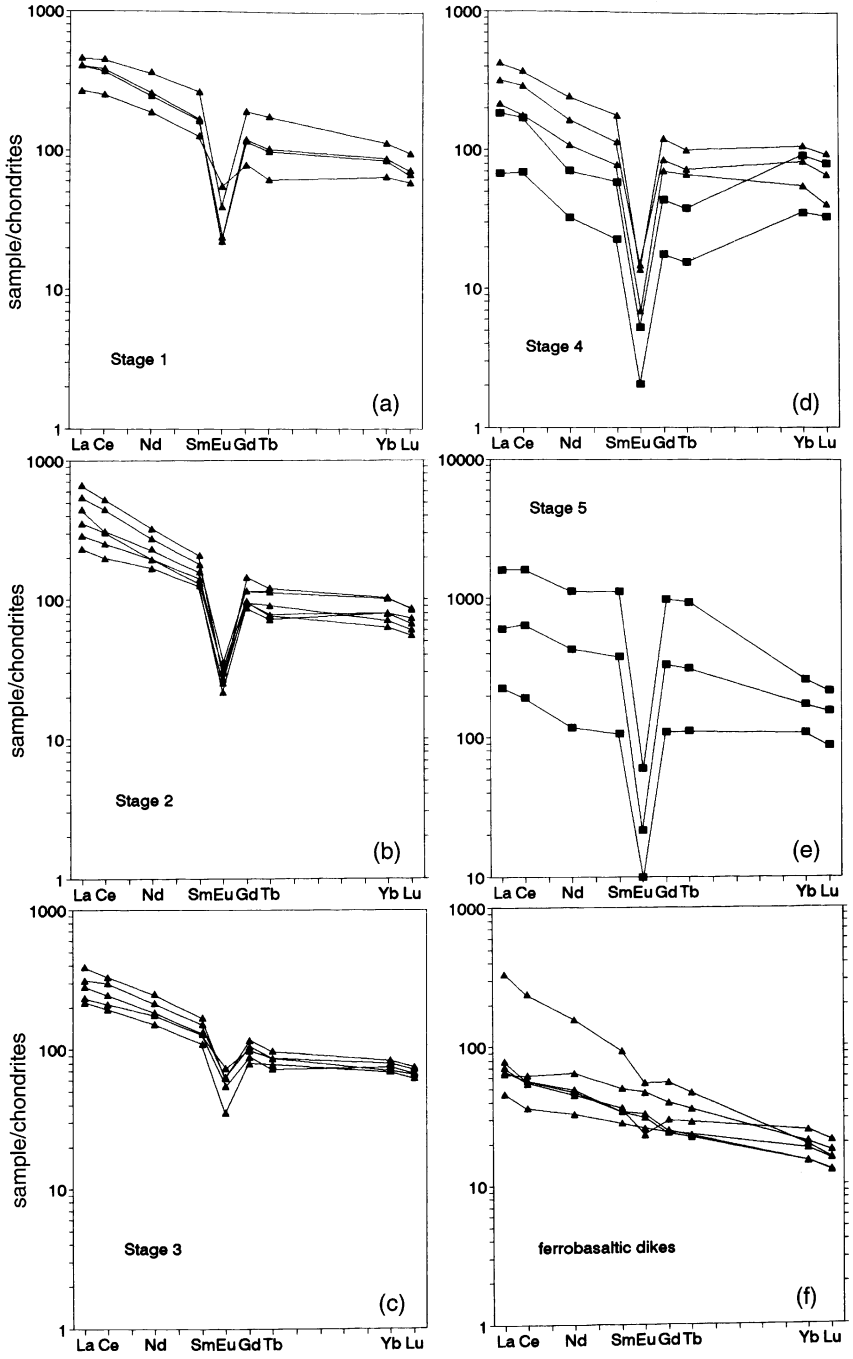


Fig. 11. Rare earth element patterns. (a) Stage 1. (b) Stage 2. (c) Stage 3. (d) Stage 4. (e) Stage 5. (f) Transitional ferrobasalt.

patterns (Sun & McDonough, 1989) but many show lower normalized Ta and Nb relative to adjacent elements. Such depletions are typical of island arc basalt (IAB; Thompson *et al.*, 1984). Such anomalies may reflect the composition of the mantle source rather than the tectonic setting.

Classification diagrams

The RBGS rocks are within-plate granites according to tectonic discrimination diagrams (Fig. 14) and are A-type or anorogenic (Loiselle & Wones, 1979; Collins *et al.*, 1982; Anderson, 1983; Clemens *et al.*, 1986; Whalen *et*

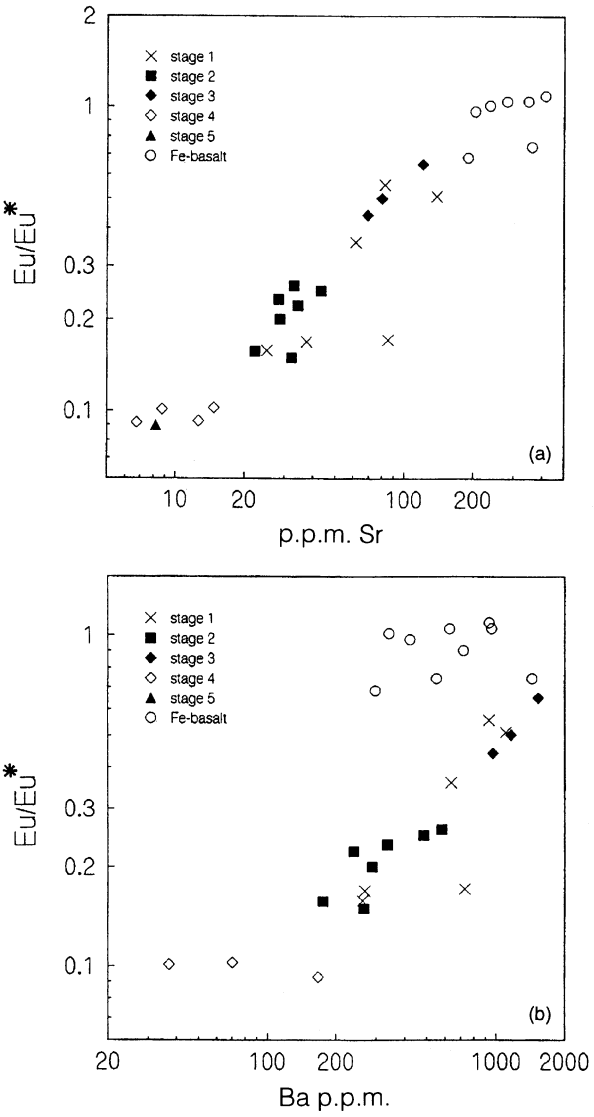


Fig. 12. Variation of the Eu anomaly (Eu/Eu*) as a function of Sr (a) and Ba (b) concentrations.

al., 1987). The origin of A-type granites has been ascribed to partial melting of crustal igneous rocks (Anderson, 1983; Creaser *et al.*, 1991), remelting of a refractory (previously melted) crustal source (Collins *et al.*, 1982; Clemens *et al.*, 1986; Whalen *et al.*, 1987), fractional crystallization of alkaline basaltic magma (Loiselle & Wones, 1979; Javoy & Weis, 1987; Eby, 1990, 1992), and fractional crystallization combined with assimilation or mixing of crustal material (Barker *et al.*, 1975; Loiselle & Wones, 1979).

Eby (1992) suggested that the Y/Nb ratio can distinguish among A-type granites with mantle (Y/Nb<1·2) or crustal (Y/Nb>1·2) origins. Most samples of stages 1, 2, 3, and 5 have Y/Nb>1·2; samples with Y/Nb<1·2 are common only among stage 4 (Fig. 14). These values imply that the suite is largely crustal in origin.

Isotopic data

Detailed discussion of isotopic compositions of the RBGS and their bearing on regional magmatic activity has been presented by Smith *et al.* (1997). In summary, data from stages 1, 4, and 5 display little variability, with low apparent initial $^{87}Sr/^{86}Sr$ (~0·7034; Norman *et al.*, 1987), positive ε_{Nd} (+ 2·5 to + 3·8; Norman *et al.*, 1987; Patchett & Ruiz, 1989). Among all five stages, $\delta^{18}O$ shows relatively little variation and low values (+ 6·8 to + 9·6; Shannon, 1994). Stage 2, the dominant exposed volume of the system, ranges from + 6·8 to + 8·0. These data along with 'mantle-like' initial Pb isotopic compositions (all five stages; Smith *et al.*, 1997) are consistent with an origin either by fractional crystallization of a mantle-derived basaltic parent or by melting of granulitic continental crust. Minor addition of a crustal component is necessary to raise $\delta^{18}O$ values. The possibility of a mantle source is thus in conflict with conclusions based on Y/Nb ratios that suggest a crustal source. In the following section, numerical models of these two possibilities are used to better constrain the origin of the suite.

PETROGENETIC MODELS

Successful models for the origin of the RBGS must explain the A-type characteristics of the suite (alkali- and Fe-rich, low compatible element concentrations, and high concentrations of HFSE and REE), the low $f(O_2)$ and high T conditions of emplacement, the mantle-like isotopic compositions, and the continuous compositional variation from quartz syenite to leucogranite (stage 3, to stage 2, to stage 4).

Partition coefficients for model calculations were taken from the literature, and are primarily derived from experimental and empirical studies of alkaline rhyolites. [A complete list of references may be obtained in Shannon (1994) or from the corresponding author.] Elements such as Zr and P are constituents of residual mineral phases so that published partition coefficient values are not strictly appropriate. For Zr and P, expressions that approximate zircon–melt and apatite–melt partition coefficients can be derived from algorithms for the zircon and apatite saturation thermometers, respectively (Watson, 1979; Watson & Harrison, 1983; Harrison & Watson, 1984). This method results in a zircon–melt partition coefficient for Zr of ~500 and an apatite–melt partitition coefficient for P of ~430.

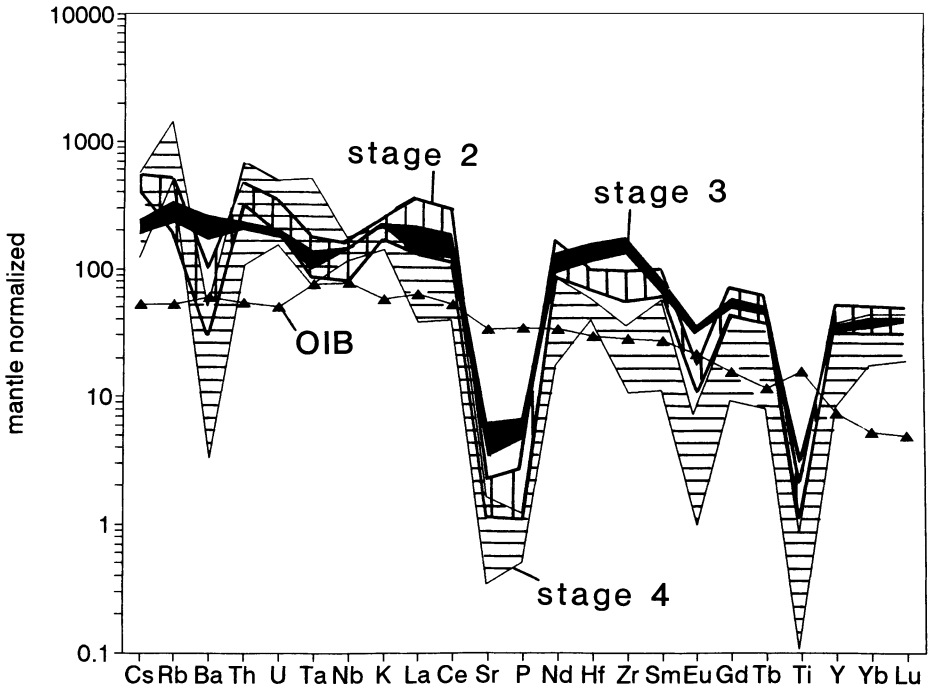


Fig. 13. Multielement diagrams for stages 2 (vertical lines), 3 (black), and 4 (horizontal lines). Normalization factors from Sun & McDonough (1989).

Fractional crystallization

Fractional crystallization was evaluated on the basis of major element variations and then tested with the trace elements. Major element mass balance calculations (Bryan *et al.*, 1969) focused on a possible liquid line of descent from parental ferrobasalt to quartz syenite of stage 3, then to granite of stage 2, and finally to leucogranite of stage 4. This sequence is not the observed order of intrusion, but it permits reasonable parent–daughter relationships for the suite to be tested. We note here that syenitic enclaves in stage 1 and stage 2 granites indicate the presence of such magmas in the system before emplacement of stage 3. Estimates of the amounts of zircon and apatite removed were made with the use of the calculated fraction of remaining liquid (F) and parent and daughter Zr and P contents.

Major element calculations show that the most mafic members of stage 3 ($SiO_2 = 61{\cdot}8\%$) can be derived from ferrobasaltic compositions after removal of ~76% crystals (Table 8). Trace element variations are generally consistent; however, rigorous testing is difficult because the large compositional gap encompasses abrupt changes in mineralogy and mineral–melt partition coefficients. In addition, most late-stage ferrobasaltic dikes are too low in Nb and Ta contents to be parental to stage 3 syenites. Because some of the ferrobasaltic dikes display appropriate parental compositions, we view this discrepancy to indicate that early, parental basaltic magmas were rarely able to reach upper-crustal levels. The petrogenesis of the ferrobasalts is the subject of continuing research (unpublished data).

Fractional crystallization from stage 3 to stage 2 was modeled in five steps (Table 8) with two types of residual mineral assemblage. In the first, sanidine, oligoclase, ferrohedenbergite, fayalite, ilmenite, and apatite were removed ('anhydrous assemblage'); in the second, sanidine, oligoclase, ferroedenite, annite, ilmenite, and apatite were removed ('hydrous assemblage'). Removal of the 'anhydrous' assemblage yielded evolved stage 2 compositions after 88% crystallization, whereas the 'hydrous' assemblage required 82% crystallization.

These models were tested against trace element variation with the Rayleigh law, and the results are plotted in Fig. 15. In the trace element tests, zircon was also considered as a fractionating phase. Neither model provides an exact fit between the actual and calculated daughter compositions. The 'anhydrous' assemblage yields a calculated daughter in which LREE, MREE, Cs, and Rb are too high, and Ba, Sr, P, Zr, Hf, Eu, Ti, Yb and Lu are too low (Fig. 15a). The 'hydrous' assemblage results in a calculated daughter that matches observed Cs, Rb, and most REE concentrations, but is too low in Ba, Sr, P, Zr, Hf, Eu, Ti, Yb, and Lu (Fig. 15).

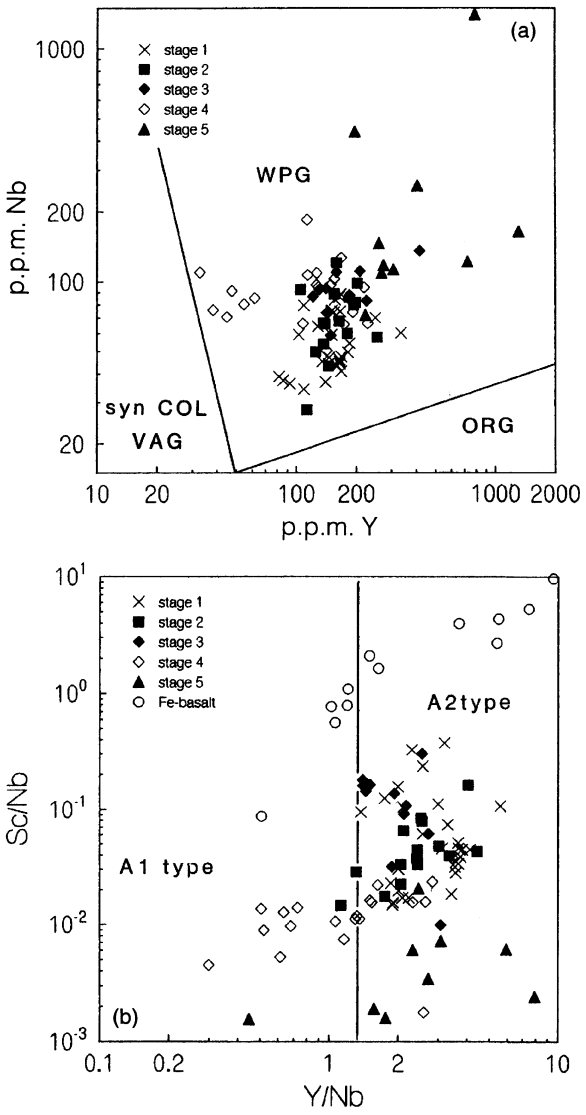


Fig. 14. (a) Tectonic discrimination diagram (Pearce *et al.*, 1984). (b) Classification of A-type granites (Eby, 1992).

Because crystal separation is probably not efficient in granitic systems, the trace element models were also calculated with the assumption that some liquid was trapped with the crystals. This '*in situ*' crystallization is modeled as a variant of Rayleigh crystallization (Langmuir, 1989) with 40% trapped liquid (Fig. 15). For the 'hydrous' assemblage model (Fig. 15b), the daughter composition plots between the '*in situ*' and the fractional crystallization models, and is generally closer to the '*in situ*' model, which suggests the presence of trapped liquid.

Fractionation from evolved stage 2 granite to stage 4 leucogranite is modeled by removal of sanidine, fayalite, annite, ilmenite, and apatite, which leaves ~9–14% residual liquid (Table 8). The calculated liquid fits the observed HFSE, but not the observed increases in LILE or the decreases in LREE and MREE (not shown). Similar results are obtained if the only mafic mineral in the assemblage is annite. Except for the LILE and Eu, addition of 0·05% of allanite and monazite to the fractionated assemblage improves the calculated trace element fit for an '*in situ*' model with 10% trapped liquid (Fig. 15c). However, these accessory minerals were not observed in thin section and their presence has yet to be confirmed by mineral separate studies. Discrepancies among the LILE and Eu may be a function of the choice of partition coefficents.

Pegmatite is rare among stage 4 dikes. This feature suggests that either H_2O-rich fluid was continually lost from stage 4 magmas during fractionation or that such fluid was not present until late in the crystallization sequence. Inasmuch as stage 4 dikes are not associated with veins or deuteric alteration, the latter idea is preferred.

Origin of stage 5

Fractional crystallization models in which stage 5 compositions are derived from evolved stage 3 syenite require a residual assemblage of sanidine, fayalite, hedenbergite, ilmenite, and apatite (Table 8). The trace element contents of daughter compositions are qualitatively well fitted by these calculations, except for elements that partition into zircon. These elements may also fit the major element model, but the proportion of residual zircon is difficult to calculate because in peralkaline melts the partition coefficient of Zr in zircon is not governed by zircon saturation equations (Watson, 1979; Watson & Harrison, 1983) and because Zr may be accepted in clinopyroxene (Duggan, 1988) and in other accessory minerals (e.g. astrophyllite; Platt & Woolley, 1986).

Stage 5 melts were probably saturated with an F-rich aqueous fluid, as suggested by their occurrence as comb-layered pegmatites with F-rich mafic minerals and by their compositional trends toward strong enrichment in HREE (Fig. 11) and Zr, Hf, Th, U, Nb, Ta, HREE, and Y (Watson, 1979; Ellison & Hess, 1986; Gwinn & Hess, 1989; Keppler, 1993). In addition, the trend of stage 5 on the pseudoternary Ab–Qz–Or plot (Fig. 5) is toward the albite apex, which is consistent with crystallization in a system that contains as much as 2–3% fluorine (Manning, 1981).

Fluorine apparently behaved as an incompatible element in RBGS magmas until fluorite became stable. Fluorite stability probably occurred at temperatures near those of biotite stability, as shown by the striking decrease in estimated F/OH from higher-T amphibole to lower-T biotite. If a melt were extracted from a deep-seated

Table 8: Results of major element mass balance calculations

Parent	Daughter	Percentage of minerals removed						*F*	Cumulative *F*	r^2
		plag†	alk. feld.	Ca-amph	biotite	ilmenite	apatite			
	Models that involve hydrous mafic minerals									
*Stage 3 syenite to stage 4 leucogranite**										
FM538 (3)	FM519 (3)	0·9	81·8	16·5	—	0·4	0·3	58·3	58	0·01
FM519 (3)	FM537 (3)	—	74·1	24·7	—	1·2	—	66·2	38	0·20
FM537 (3)	FM157 (2)	—	76·7	22·3	—	0·9	—	61·8	24	0·07
FM157 (2)	FM133 (2)	—	76·7	22·4	0·7	0·2	—	78·4	19	0·09
FM133 (2)	FM136 (4)	—	87·9	—	10·8	0·2	1·1	77·9	14	0·25
	Models that involve anhydrous mafic minerals									
		plag†	alk. feld.	olivine	cpx	ilmenite	apatite			
Ferrobasalt to stage 3 syenite (using forsteritic olivine and magnesian cpx)										
FM9401	FM538 (3)	57·4	—	23·0	7·6	9·7	2·2	21·7	—	0·02
Stage 3 syenite to stage 4 leucogranite (using fayalitic olivine and ferrohedenbergite)										
FM538 (3)	FM519 (3)	—	85·7	4·2	9·2	0·9	0·1	43·3	43	0·01
FM519 (3)	FM537 (3)	—	83·3	7·1	7·5	1·6	0·6	55·6	24	0·10
FM537 (3)	FM801 (3)	—	80·5	5·7	12·3	1·2	0·3	69·8	17	0·02
FM801 (3)	FM157 (2)	—	90·9	8·0	—	1·1	—	82·9	14	0·02
FM157 (2)	FM133 (2)	—	85·9	11·9	—	0·6	1·6	80·0	11	0·09
FM133 (2)	FM136 (4)	—	92·7	5·6	—	1·0	0·8	77·6	9	0·47
Stage 3 syenite to stage 5 peralkaline granite										
FM801	FM623	—	89·5	5·5	3·9	0·9	0·2	47·0	—	0·10
FM801	FMMK1	—	95·3	2·4	1·2	0·8	0·3	45·6	—	0·11

*Sample numbers followed by stage number in parentheses.
†Plagioclase composition An_{52}.

crystal mush of stage 2 composition before biotite and fluorite stability and then raised to shallow levels in the crust (e.g. the depth of emplacement; ~1–2 km), it might reach fluid saturation because of the decrease in confining pressure. If so, fluorine would partition preferentially into the melt, relative to H_2O (e.g. Bailey, 1977) and would drive the melt to more F-rich and HFSE-rich compositions. This implies a deeper-seated origin of stage 5 magmas, which is consistent with the long strike length of stage 5 dikes, as much as 1·5 km.

Partial melting

Several possible crustal source rocks are present in the basement of west Texas and adjacent areas. The most likely sources are mafic, charnockitic, and granodioritic orthogneiss and pelitic paragneiss, which make up part of the xenolith suite at nearby Kilbourne Hole (Fig. 1b; Padovani & Carter, 1977). Inasmuch as melting of a pelitic source would produce peraluminous melts, pelitic paragneiss is not considered further. The other xenolith types were used as possible sources in equilibrium fusion models in which residual mineral assemblages and proportions were taken from experimental studies (Conrad *et al.*, 1988; Rushmer, 1991, 1993; Beard & Lofgren, 1991; Wolf & Wyllie, 1991; Skjerlie & Johnston, 1992, 1993; Beard *et al.*, 1993; Skjerlie *et al.*, 1993). The work of Beard *et al.* (1993) was particularly useful because their experiments were done on Kilbourne Hole xenoliths.

Granodioritic or tonalitic sources can produce granitic melts that generally have intermediate *mg*-number and Ca contents (Skjerlie & Johnston, 1992, 1993), unlike those of the RBGS. For reasonable degrees of melting, calculated HFSE (especially Nb and Ta) and HREE compositions

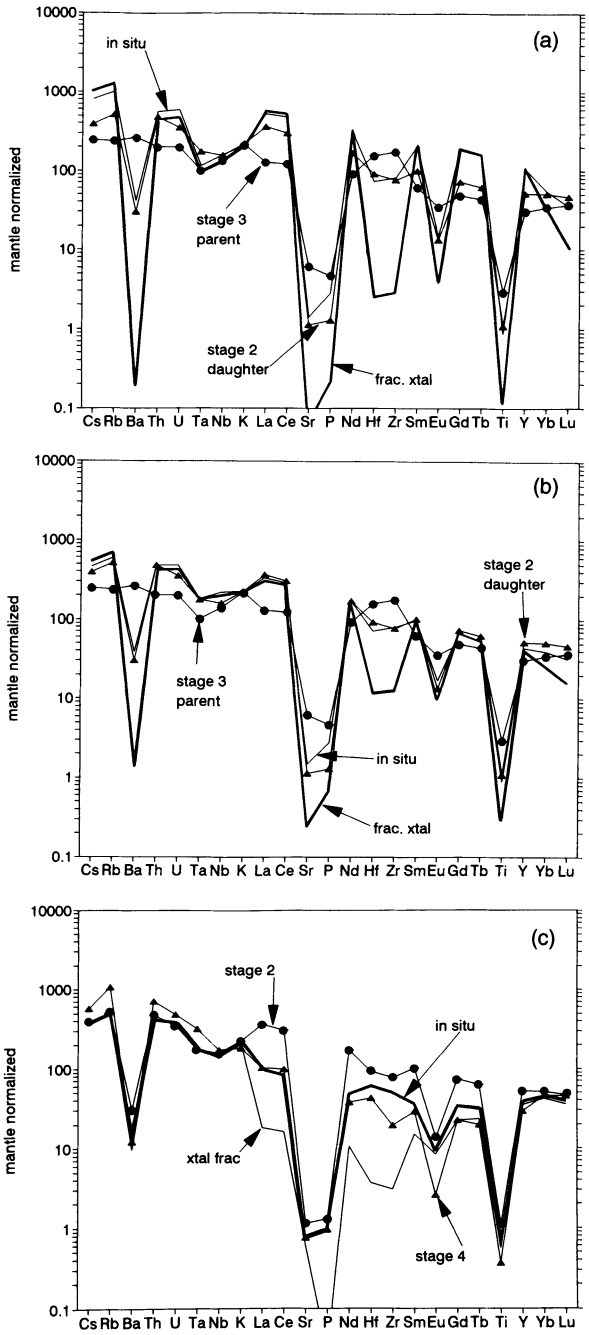


Fig. 15. Mantle-normalized multielement plots that show the results of trace element tests of fractional crystallization, *in situ* crystallization, and melting models. (a) Crystallization model from stage 3 quartz syenitic parent to stage 2 granite in which anhydrous mafic minerals are part of the fractionated assemblage. '*In situ*' crystallization indicates that removal of crystals was inefficient (see text). (b) Crystallization model as in (a) except that hydrous mafic minerals are part of the fractionated assemblage. (c) Crystallization model from stage 2 granitic parent to stage 4 leucogranite (see text).

of melts from granodioritic and tonalitic sources are low relative to the RBGS. Similarly, partial melting of mafic orthogneiss typically produces calc-alkaline or I-type rather than A-type granitoids (Rushmer, 1991; Beard & Lofgren, 1991; Wolf & Wyllie, 1991).

The best results of partial melting models used a charnockitic source even though the results generally provide a poor calculated fit to the LILE and calculated abundances of HFSE, HREE, and Y are lower than observed. Acceptable fits to these trace elements can only be achieved when a hypothetical charnockitic source with anomalously high HREE, U, and Th is used. However, few charnockitic rocks have the necessary HREE enrichment (e.g. Condie & Allen, 1984).

Mafic alkaline volcanic rocks are probably not appropriate sources because subaerial exposure would result in oxidation of the volcanic rocks, and subsequent partial melts would display high $f(O_2)$ and low Fe/Mg, unlike the RBGS. Small degrees of partial melting (<5%) of an amphibole-bearing alkali gabbro could result in major element compositions similar to stage 2 granites (but see below). However, large degrees of partial melting are required to progress from stage 2 to stage 3 compositions. In such models, the residual minerals necessary to produce low Ba, Nb, P, Sr, Zr, Eu, and Ti of the most differentiated stage 2 rocks result in calculated abundances of HFSE and HREE lower than those observed.

Any partial melting model should be able to explain the progressive depletions of Ba, Sr, P, Zr, Hf, Eu, and Ti from stage 3 to stage 2 to stage 4 (Fig. 13). These trends suggest a residual mineral assemblage that contains zircon, alkali feldspar ± plagioclase, apatite, and Ti-rich minerals. Bulk partition coefficients necessary to achieve these depletions are in the range of 2–4 for Ti, Zr, Sr, Ba, and P. Melting models in which zircon, apatite, ilmenite, and amphibole are residual phases (such as amphibole-bearing alkali gabbro) can reproduce trace element patterns of individual rocks, but fail to produce the trace element trends observed in the suite. This effect is illustrated in Fig. 16, in which partial melting paths are compared with fractional and *in situ* crystallization paths. Patterns similar to those shown in Fig. 16 are also apparent in plots of Rb vs Ba.

DISCUSSION

Melting models indicate that special conditions [high T, low $f(O_2)$] and compositions ('enriched' charnockite) are necessary to produce magmas such as those parental to the RBGS from a crustal source. However, fractional crystallization from mantle-derived, mildly alkaline basaltic parents meets the high T and low $f(O_2)$ requirements and the mantle-like isotopic compositions, and can successfully fit much of the major and trace element variation

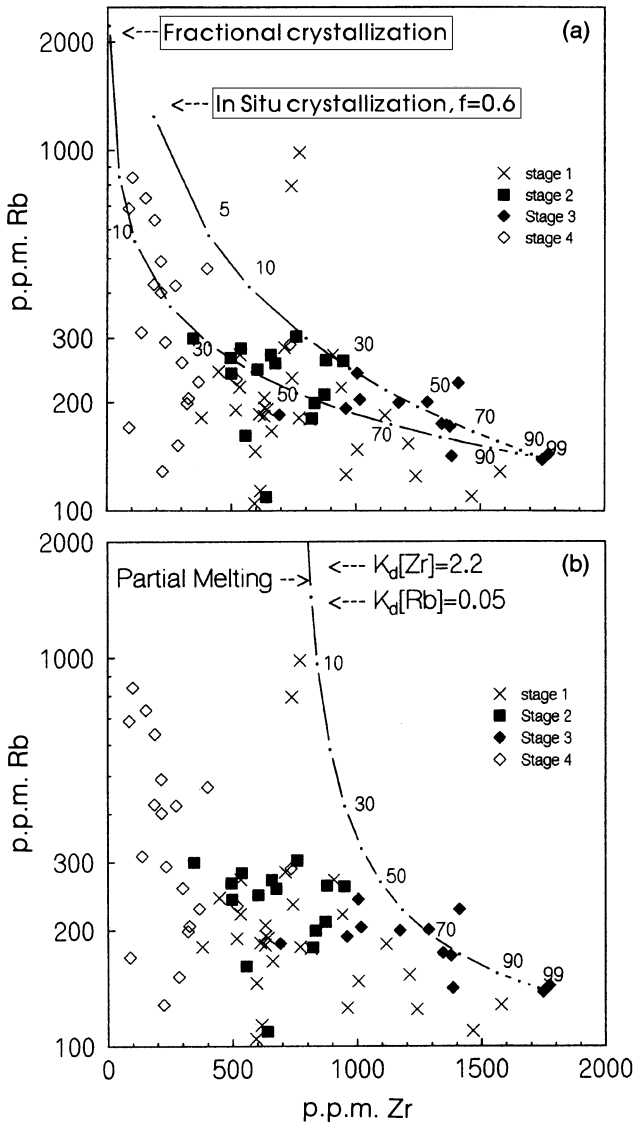


Fig. 16. Plots of Zr and Rb variation compared with crystallization and partial melting models. For fractional crystallization and *in situ* crystallization models (a), the bulk partition coefficients were calculated from mass balance models to be 0·4 for Rb and 2·2 for Zr. The *in situ* crystallization model assumes 60% of the melt was extracted. For the melting models (b), bulk partition coefficients used were 0·05 for Rb and 2·2 for Zr. The lower value for Rb assumes no residual micas in the source, as is consistent with the metaluminous composition for the bulk of the RBGS.

from a basaltic parent to stage 4 leucogranite. The high Y/Nb ratios of stages 1–3, which are suggestive of a crustal source (Eby, 1992), are in this case probably a reflection of Nb depletion in the mantle source of the parental basaltic magmas (Condie, 1992).

The crystallization model requires extensive fractionation of large volumes of basaltic parent, with subsequent high-density cumulates. Gravity models of the Franklin Mountains indicate the presence of a large positive anomaly beneath the northwestern part of the range (G. R. Keller, personal communication, 1993). This anomaly could represent some of the cumulate material associated with the RBGS. A significant amount of the crystallization of parental basaltic magmas could have occurred near the Moho, in which case no gravity anomaly would be apparent. An origin by fractional crystallization also means that the RBGS and related cumulate rocks are juvenile additions to the continental crust during Grenville time.

Proposed tectonic settings of RBGS magmatism have encompassed arc, back-arc, or continental rift-like environments. Our data are not consistent with an arc

setting and no compelling geochemical evidence exists for an arc signature. A back-arc setting is more difficult to evaluate, in that a range of parental magmas is possible. We tend to discount a back-arc setting for two reasons. First, many back-arc magmas display geochemical features that suggest a subduction-related origin (e.g. Sinton & Fryer, 1987). Except for the low Nb and Ta in ferrobasaltic dikes, such features are lacking in the RBGS. Second, if the RBGS formed in a back-arc setting, the associated arc must have been located to the south or southeast. In the Van Horn area (Fig. 1b), the clastic sedimentary unit related to Grenville deformation (the Hazel Formation) contains leucogranite clasts that were apparently derived from a southerly source (Soegaard & Callahan, 1993, 1994). These clasts are compositionally similar to stage 4 of the RBGS (Barnes & Shannon, unpublished data 1994), which suggests that A-type plutons similar to the Red Bluff suite (and not to arc-related granites) exist in the subsurface south of the Franklin Mountains.

The RBGS is slightly younger than a large, 1·16–1·07 Ga layered mafic complex, the Nellie intrusion (Keller *et al.*, 1989), which occurs in the subsurface east of the Franklin Mountains (Fig. 1b). This intrusion is predominantly noritic with sparse peridotitic cumulates (Hoover *et al.*, 1985; Kargi & Barnes, 1995). The shape of its residual gravity anomaly (Keller *et al.*, 1989; Adams & Keller, 1994), its similarity in age and trend to the Mid-Continent Rift (Keller *et al.*, 1989), its predominantly noritic composition (Hoover *et al.*, 1985; Kargi & Barnes, 1995), and the compositions of its primary minerals (Kargi & Barnes, 1995) all suggest emplacement in an extensional environment, and probably in a rift. An extensional environment would also be appropriate for the RBGS, with its petrologic evidence for mildly alkaline parental basalts. It is noteworthy that the setting of the RBGS (to the west of a continental rift) is similar to that of the nearly coeval Pikes Peak batholith, which was emplaced to the west of the Mid-Continent Rift. The RBGS is mineralogically and compositionally nearly identical to the sodic series of the Pikes Peak batholith (Barker *et al.*, 1975).

CONCLUSIONS

The Red Bluff granitic suite consists of a compositionally continuous series of A-type quartz syenite and granite, with late-stage peralkaline pegmatitic dikes. Late-stage transitional ferrobasaltic dikes attest to the presence of mantle-derived magmas late, and possibly throughout, the history of the suite. The granitic magmas were iron rich, had low $f(O_2)$, low initial H_2O contents, and moderate fluorine contents. The least evolved syenitic magmas were emplaced at temperatures near 1050°C.

Numerical models are consistent with inefficient fractional crystallization as the cause of chemical variation from quartz syenitic to leucogranitic compositions. They also indicate that the quartz syenitic magmas could result from fractional crystallization of a mantle-derived basaltic parent. Late-stage peralkaline dikes may represent F-rich residual liquids tapped from deep levels of the crystallizing system.

An origin by partial melting of known Proterozoic crustal rocks is not compatible with element concentrations or trends in the RBGS. Thus, the suite represents an addition to the continental crust during Grenville time.

Trace element patterns and modeling indicate that the parental magmas of the RBGS were mildly alkaline and OIB-like, rather than arc-like. When combined with regional geophysical and magmatic patterns, these features suggest that the suite formed in an extensional environment.

ACKNOWLEDGEMENTS

We wish to thank M. A. Barnes for assistance in the laboratory, E. Y. Anthony, J. D. Hoover, G. R. Keller, P. Goodell, K. Johnson, and J. McMurry for useful discussions on the petrology of the suite, and J. L. Anderson, B. Nelson, and J. M. Rhodes for helpful comments during review. Samples were collected under Franklin Mountain State Park permit 43-89. This research was supported by the donors of the Petroleum Research Fund of the American Chemical Society; INAA analyses were provided by the DOE Reactor Sharing Program at Oregon State University.

REFERENCES

Abrecht, J. & Hewitt, D. A., 1988. Experimental evidence on the substitution of Ti in biotite. *American Mineralogist* **73,** 1275–1284.

Adams, D. C. & Keller, G. R., 1994. Possible extension of the Mid-continent Rift in west Texas and eastern New Mexico. *Canadian Journal of Earth Sciences* **31,** 709–720.

Anderson, J. L., 1980. Mineral equilibria and crystallization conditions in the late Precambrian Wolf River rapakivi massif, Wisconsin. *American Journal of Science* **280,** 289–332.

Anderson, J. L., 1983. Proterozoic anorogenic granite plutonism of North America. *Geological Society of America Memoir* **161,** 133–154.

Anderson, J. L. & Bender, E. E., 1989. Nature and origin of Proterozoic A-type granitic magmatism in the southwestern United States of America. *Lithos* **23,** 19–52.

Anthony, E. Y., Barnes, C. G., Chen, W., Hoffer, J. M., Keller, G. R., Marsaglia, K. M., McLemore, V. T., Seeley, J. M., Seward, M. A., Shannon, W. M. & Thomann, W. F., 1991. Examples of modern rift volcanism and Proterozoic anorogenic magmatism: the Potrillo Volcanic Field of southern New Mexico and the Franklin Mountains of west Texas. *New Mexico Bureau of Mines and Mineral Resources Bulletin* **137,** 1–3.

Bailey, J. C., 1977. Fluorine in granitic rocks and melts: a review. *Chemical Geology* **19,** 1–42.

Barker, F., Wones, D. R., Sharp, W. N. & Desborough, G. A., 1975. The Pikes Peak batholith, Colorado Front Range, and a model for the origin of the gabbro–anorthosite–syenite–potassic granite suite. *Precambrian Research* **2,** 97–160.

Beard, J. S. & Lofgren, G. E., 1991. Dehydration melting and water-saturated melting of basaltic and andesitic greenstones and amphibolites at 1, 3, and 6·9 kb. *Journal of Petrology* **32,** 365–401.

Beard, J. S., Abitz, R. J. & Lofgren, G. E., 1993. Experimental melting of crustal xenoliths from Kilbourne Hole, New Mexico and implications for the contamination and genesis of magmas. *Contributions to Mineralogy and Petrology* **115,** 88–102.

Bence, A. E. & Albee, A. L., 1968. Empirical correction factors for the electron microanalysis of silicates and oxides. *Journal of Geology* **76,** 382–402.

Bonin, B., 1986. *Ring Complex Granites and Anorogenic Magmatism*. New York: Elsevier, 188 pp.

Bonin, B. & Giret, A., 1985. Clinopyroxene compositional trends in oversaturated and undersaturated alkaline ring complexes. *Journal of African Earth Sciences* **3,** 175–183.

Bryan, W. B., Finger, L. W. & Chayes, F., 1969. A least squares approximation for estimating the composition of a mixture. *Carnegie Institution of Washington, Yearbook* **67,** 244–247.

Caruba, R., Baumer, A., Ganteaume, M. & Iacconi, P., 1985. An experimental study of hydroxyl groups and water in synthetic and natural zircons: a model of the metamict state. *American Mineralogist* **70,** 1224–1231.

Clemens, J. D., Holloway, J. R. & White, A. J. R., 1986. Origin of an A-type granite: experimental constraints. *American Mineralogist* **71,** 317–324.

Collins, W. J., Beams, S. D., White, A. J. R. & Chappell, B. W., 1982. Nature and origin of A-type granites with particular reference to southeastern Australia. *Contributions to Mineralogy and Petrology* **80,** 189–200.

Condie, K. C., 1982. Plate tectonics model for Proterozoic continental accretion in the southwestern United States. *Geology* **10,** 37–42.

Condie, K. C., 1992. Proterozoic terranes and continental accretion in southwestern North America. In: Condie, K. C. (ed.) *Proterozoic Crustal Evolution, Developments in Precambrian Geology 10*. Amsterdam: Elsevier, pp. 447–480.

Condie, K. C. & Allen, P., 1984. Origin of Archean charnockites from southern India. In: Kroner, A., Hanson, G. N. & Goodwin, A. M. (eds) *Archean Geochemistry*. Berlin: Springer-Verlag, pp. 182–203.

Condie, K. C. & Budding, A. J., 1979. Geology and geochemistry of Precambrian rocks, central and south–central New Mexico. *New Mexico Bureau of Mines and Mineral Resources Memoir* **35,** 58 pp.

Conrad, W. K., Nicholls, I. A. & Wall, V. J., 1988. Water-saturated and -undersaturated melting of metaluminous and peraluminous crustal compositions at 10 kb: evidence for the origin of silicic magmas in the Taupo volcanic zone, New Zealand, and other occurrences. *Journal of Petrology* **29,** 765–803.

Copeland, P. & Bowring, S. A., 1988. U–Pb zircon and $^{40}Ar/^{39}Ar$ ages from Proterozoic rocks, west Texas. *Geological Society of America Abstracts with Programs* **20,** 95–96.

Creaser, R. A., Price, R. C. & Wormald, R. J., 1991. A-type granites revisited: assessment of a residual-source model. *Geology* **19,** 163–166.

Czamanske, G. K. & Dillet, B., 1988. Alkali amphibole, tetrasilicic mica, and sodic pyroxene in peralkaline siliceous rocks, Questa caldera, New Mexico. *American Journal of Science* **288-A,** 358–392.

Deer, W. A., Howie, R. A. & Zussman, J., 1966. *An Introduction to the Rock Forming Minerals*. New York: John Wiley, 528 pp.

Duggan, M. B., 1988. Zirconium-rich sodic pyroxenes in felsic volcanics from Warrumbungle volcano, central New South Wales, Australia. *Mineralogical Magazine* **52,** 491–496.

Eby, G. N., 1990. The A-type granitoids: a review of their occurrence and chemical characteristics and speculations on their petrogenesis. *Lithos* **26,** 115–134.

Eby, G. N., 1992. Chemical subdivision of the A-type granitoids: petrogenetic and tectonic implications. *Geology* **20,** 641–644.

Ellison, A. J. & Hess, P. C., 1986. Solution behavior of + 4 cations in high silica melts: petrologic and geochemical implications. *Contributions to Mineralogy and Petrology* **94,** 343–351.

Ernst, W. G., 1962. Synthesis, stability relations, and occurrence of riebeckite and riebeckite–arfvedsonite solid solutions. *Journal of Geology* **70,** 689–736.

Gilbert, M. C., Helz, R. T., Popp, R. K. & Spear, F. S., 1982. Experimental studies of amphibole stability. In: Veblen, D. R. & Ribbe, P. H. (eds) *Amphiboles: Petrology and Experimental Phase Relations. Mineralogical Society of America, Reviews in Mineralogy* **9A,** 229–353.

Giret, A., Bonin, B. & Leger, J.-M., 1980. Amphibole compositional trends in oversaturated and undersaturated alkaline plutonic ring-complexes. *Canadian Mineralogist* **18,** 481–495.

Guidotti, C. V., 1984. Micas in metamorphic rocks. In: Bailey, S. W. (ed.) *Micas. Mineralogical Society of America, Reviews in Mineralogy* **13,** 357–467.

Gwinn, R. & Hess, P. C., 1989. Iron and titanium solution properties in peraluminous and peralkaline rhyolitic liquids. *Contributions to Mineralogy and Petrology* **101,** 326–338.

Hammond, J. G., 1990. Middle Proterozoic diabase intrusions in the southwestern U.S.A., as indicators of limited extensional tectonism. In: Gowers, C. F. (ed.) *Mid-Proterozoic Laurentia–Baltica. Geological Association of Canada, Special Paper* **38,** 517–531.

Harbour, R. L., 1960. Precambrian rocks at North Franklin Mountain, Texas. *Bulletin, American Association of Petroleum Geologists* **44,** 1785–1792.

Harbour, R. L., 1972. Geology of the Northern Franklin Mountains, Texas and New Mexico. *US Geological Survey Bulletin* **1298,** 129 pp.

Harrison, T. M. & Watson, E. B., 1984. The behavior of apatite during crustal anatexis: equilibrium and kinetic considerations. *Geochimica et Cosmochimica Acta* **48,** 1467–1477.

Hawthorne, F. C., 1983. The crystal chemistry of amphiboles. *Canadian Mineralogist* **21,** 173–480.

Hoffer, R. L., 1976. Contact metamorphism of the Precambrian Castner Marble, Franklin Mountains, El Paso County. M.S. Thesis, University of Texas at El Paso, 77 pp.

Hoover, J. D., Keller, G. R. & Hills, J. M., 1985. The Nellie intrusion: a basic stratiform intrusion in the Central basin platform in west Texas. *EOS* **66,** 1136.

Irvine, T. N. & Baragar, W. R. A., 1971. A guide to the chemical classification of the common volcanic rocks. *Canadian Journal of Earth Sciences* **8,** 523–548.

Javoy, M. & Weis, D., 1987. Oxygen isotopic composition of alkaline anorogenic granites as a clue to their origin: the problem of crustal oxygen. *Earth and Planetary Science Letters* **84,** 415–422.

Kargi, H. & Barnes, C. G., 1995. A Grenville-age layered intrusion in the subsurface of west Texas: petrology, petrography, and possible tectonic setting. *Precambrian Research* **32,** 2159–2166.

Keller, G. R., Hills, J. M., Baker, M. R. & Wallin, E. T., 1989. The Nellie intrusion: a basic stratiform intrusion in the Central basin platform in west Texas. *Geology* **17,** 1049–1052.

Keppler, H., 1993. Influence of fluorine on the enrichment of high field strength trace elements in granitic rocks. *Contributions to Mineralogy and Petrology* **114,** 479–488.

Langmuir, C. H., 1989. Geochemical consequences of *in situ* crystallization. *Nature* **340,** 199–205.

Larson, E. E., Patterson, P. E. & Mutschler, F. E., 1994. Lithology, chemistry, age, and origin of the Proterozoic Cardenas Basalt, Grand Canyon, Arizona. *Precambrian Research* **65,** 255–276.

Leake, B.E., 1978. Nomenclature of amphiboles. *Canadian Minereralogist* **16,** 501–520.

Loiselle, M. G. & Wones, D. R., 1979. Characteristics and origins of anorogenic granite. *Geological Society of America, Abstracts with Programs* **11,** 468.

Lovejoy, E. M. P., 1975. An interpretation of the structural geology of the Franklin Mountains, Texas. *New Mexico Geological Society Guidebook, 26th Field Conference.*, Socorro: New Mexico Geological Society, pp. 261–268.

Ludwig, K. R., 1980. Calculation of uncertainties of U–Pb isotopic age data. *Earth and Planetary Science Letters* **46,** 212–220.

Ludwig, K. R., 1983. Plotting and regression programs for isotope geochemists, for use with HP-86/87 microcomputers. *US Geological Survey Open-File Report* **83-849**, 94 pp.

Luth, W. C., Jahns, R. H. & Tuttle, O. F., 1964. The granite system at pressures of 4 to 10 kilobars. *Journal of Geophysical Research* **69,** 759–773.

Manning, D. A. C., 1981. The effect of fluorine on liquidus phase relationships in the system Qz–Ab–Or with excess water at 1 kb. *Contributions to Mineralogy and Petrology* **76,** 206–215.

Mosher, S., 1993. Proterozoic rocks east and southeast of the Grenville front. In: Reed, J. C., Jr, Bickford, M. E., Houston, R. S., Link, P. K., Rankin, D. W., Sims, P. K. & Van Schmus, W. R. (eds) *The Geology of North America* **C-2,** *Precambrian: Conterminous U.S.* Boulder, CO: Geological Society of America, pp. 365–378.

Munoz, J. L., 1969. Experimental control of fluorine reactions in hydrothermal systems. *American Mineralogist* **54,** 943–959.

Munoz, J. L., 1984. F–OH and Cl–OH exchange in micas with applications to hydrothermal ore deposits. In: Bailey, S. W. (ed.) *Micas. Mineralogical Society of America, Reviews in Mineralogy* **13,** 469–494.

Nekvasil, H., 1990. Reaction relations in the granite system: implications for trachytic and syenitic magmas. *American Mineralogist* **75,** 560–571.

Nelson, B. K. & DePaolo, D. J., 1985. Rapid production of continental crust 1·7 to 1·9 b.y. ago: Nd isotopic evidence from the basement of the North American mid-continent. *Geological Society of America Bulletin* **96,** 746–754.

Norman, D. I., Condie, K. C., Smith, R. W. & Thomann, W. F., 1987. Geochemical and Sr and Nd isotopic constraints on the origin of late Proterozoic volcanics and associated tin-bearing granites from the Franklin Mountains, west Texas. *Canadian Journal of Earth Sciences* **24,** 830–838.

Padovani, E. R. & Carter, J. L., 1977. Aspects of the deep crustal evolution beneath south central New Mexico. In: Heacock, J. G. (ed.) *The Earth's Crust*. Washington, DC: American Geophysical Union, pp. 19–55.

Patchett, P. J. & Ruiz, J., 1989. Nd isotopes and the origin of Grenville-age rocks in Texas: implications for Proterozoic evolution of the United States mid-continent region. *Journal of Geology* **97,** 685–695.

Pearce, J. A., Harris, N. B. W. & Tindle, A. G., 1984. Trace element discrimination diagrams for the tectonic interpretation of granitic rocks. *Journal of Petrology* **25,** 956–983.

Pichavant, M. & Manning, D. A. C., 1984. Petrogenesis of tourmaline granites and topaz granites; the contribution of experimental data. *Physics of Earth and Planetary Interiors* **35,** 31–50.

Pittenger, M. A., Marsaglia, K. M. & Bickford, M. E., 1994. Depositional history of the Middle Proterozoic Castner Marble and basal Mundy Breccia, Franklin Mountains, west Texas. *Journal of Sedimentary Research* **B64,** 282–297.

Platt, G. A. & Woolley, A. R., 1986. The mafic mineralogy of the peralkaline syenites and granites of the Mulanje complex, Malawi. *Mineralogical Magazine* **50,** 85–99.

Ray, R. D., 1982. Geology of the Precambrian Red Bluff granite complex, Fullelman canyon area, Franklin Mountains, El Paso County, Texas. M.S. Thesis, University of Texas at El Paso, 295 pp.

Roths, P., 1991. Geology of Proterozoic outcrops in Dead Man and Little San Niclolas Canyons, southern San Andres Mountains, New Mexico. *New Mexico Geological Society Guidebook, 42nd Field Conference.*, Socorro: New Mexico Geological Society, pp. 91–96.

Roths, P., 1993. Geochemical and geochronological studies of the Grenville-age (1250–1000 Ma) Allamore and Hazel Formations, Hudspeth and Culberson Counties, West Texas. *Geological Society of America, South–Central Section, Field Trip Guidebook*. Dallas: University of Texas, pp. 11–35.

Rushmer, T., 1991. Partial melting of two amphibolites: contrasting experimental results under fluid-absent conditions. *Contributions to Mineralogy and Petrology* **107,** 41–59.

Rushmer, T., 1993. Experimental high-pressure granulites: some applications to natural mafic xenolith suites and Archean granulite terranes. *Geology* **21,** 411–414.

Shannon, W. M., 1994. Petrogenesis of Middle Proterozoic alkaline granitic rocks, Franklin Mountains, West Texas. Ph.D. Thesis, Texas Tech University, Lubbock, 240 pp.

Sinton, J. M. & Fryer, P., 1987. Mariana trough lavas from 18°N: implications for the origin of back arc basin basalts. *Journal of Geophysical Research* **92,** 12782–12802.

Skjerlie, K. P. & Johnston, A. D., 1992. Vapor-absent melting at 10 kbar of a biotite- and amphibole-bearing tonalitic gneiss: implications for the generation of A-type granites. *Geology* **20,** 263–266.

Skjerlie, K. P. & Johnston, A. D., 1993. Fluid absent melting behavior of an F-rich tonalitic gneiss at mid-crustal pressures: implications for the generation of anorogenic granites. *Journal of Petrology* **34,** 785–815.

Skjerlie, K. P., Patiño Douce, A. E. & Johnston, A., 1993. Fluid absent melting of a layered crustal protolith: implications for the generation of anatectic granites. *Contributions to Mineralogy and Petrology* **114,** 365–378.

Smith, D. R., Barnes, C., Shannon, W., Roback, R. & James, E., 1997. Mid-Proterozoic granitic magmatism in Texas. *Precambrian Research*. In press.

Soegaard, K. & Callahan, D. M., 1993. Sedimentology of Hazel Formation. In: Soegaard, K., Nielsen, K. C., Marsaglia, K. & Barnes, C. (eds) *Precambrian Geology of Franklin Mountains and Van Horn Area, Trans-Pecos, Texas. Geological Society of America, South Central Section Field Trip Guidebook*. Dallas: University of Texas, pp. 49–62.

Soegaard, K. & Callahan, D. M., 1994. Late Middle Proterozoic Hazel Formation near Van Horn, Trans-Pecos, Texas: evidence for transpressional deformation in Grenville basement. *Geological Society of America Bulletin* **106,** 413–423.

Spear, F. S. & Kimball, K. L., 1984. RECAMP—a FORTRAN IV program for estimating Fe^{3+} contents in amphiboles. *Computers and Geosciences* **10,** 317–325.

Stacey, J. S. & Kramers, J. D., 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. *Earth and Planetary Science Letters* **26,** 207–221.

Steiger, R. & Jaeger, E., 1977. Subcommittee on geochronology: convention on the use of decay constants in geo- and cosmochronology. *Earth and Planetary Science Letters* **36,** 359–362.

Streckeisen, A., 1974. Classification and nomenclature of plutonic rocks: recommendations of the IUGS Subcommission on the Systematic of Igneous Rocks. *Geologische Rundschau* **63,** 773–786.

Sun, S.-S. & McDonough, W. F., 1989. Chemical and isotopic systematics of oceanic basalts: implications for mantle composition and

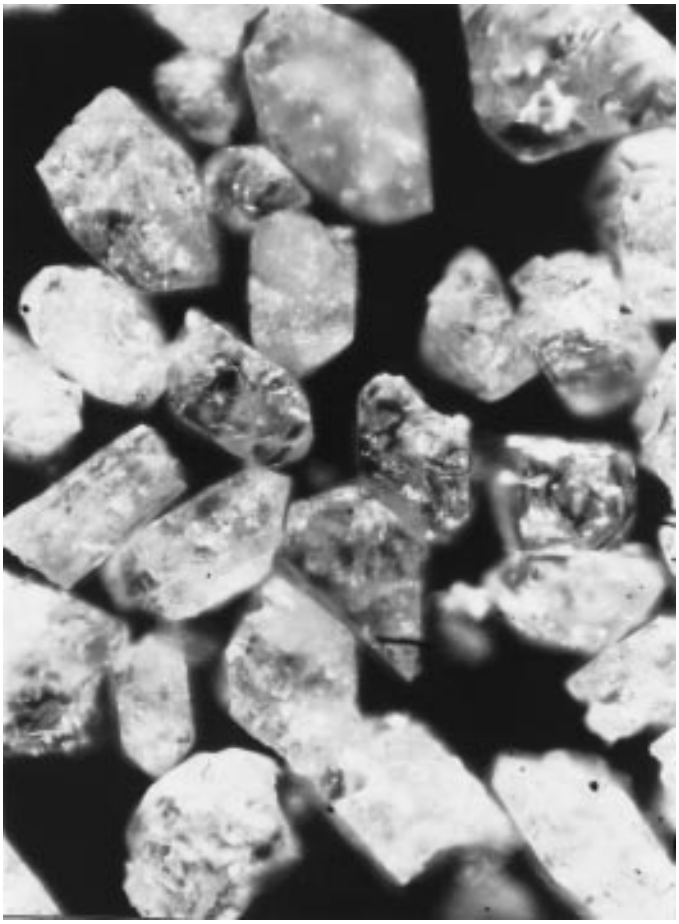


Fig. A1. Incident light photograph of non-magnetic fraction of zircons from sample FM 157 of stage 2 granite. 50 × magnification.

process. In: Saunders, A. D. & Norry, M. J. (eds) *Magmatism in the Ocean Basins. Geological Society Special Publication* **42,** 313–345.

Thomann, W. F., 1980. Ignimbrites, trachytes and sedimentary rocks of the Precambrian Thunderbird Group, Franklin Mountains, El Paso, Texas. *Geological Society of America Bulletin* **92,** 94–100.

Thompson, R. N., Morrison, M. A., Hendry, G. L. & Parry, S. J., 1984. An assessment of the relative roles of crust and mantle in magma genesis: an elemental approach. *Philosophical Transactions of the Royal Society of London, Series A* **310,** 549–590.

Tuttle, O. F. & Bowen, N. L., 1958. Origin of granite in the light of experimental studies in the system $NaAlSi_3O_8$–$KAlSi_3O_8$–SiO_2–H_2O. *Geological Society of America Memoir* **74,** 153 pp.

Wasserburg, G. J., Wetherill, G. W., Silver, L. T. & Flawn, P. T., 1962. A study of the ages of the Precambrian of Texas. *Journal of Geophysical Research* **67,** 4021–4047.

Watson, E. B., 1979. Zircon saturation in felsic liquids: experimental results and applications to trace element geochemistry. *Contributions to Mineralogy and Petrology* **70,** 407–419.

Watson, E. B. & Harrison, T. M., 1983. Zircon saturation revisited: temperature and composition effects in a variety of crustal magma types. *Earth and Planetary Science Letters* **64,** 295–304.

Whalen, J. B., Currie, K. L. & Chappell, B. W., 1987. A-type granites: geochemical characteristics, discrimination and petrogenesis. *Contributions to Mineralogy and Petrology* **95,** 407–419.

Wolf, M. B. & Wyllie, P. J., 1991. Dehydration-melting of solid amphibolite at 10 kbar: textural development, liquid interconnectivity and applications to the segregation of magmas. *Mineralogy and Petrology* **44,** 151–179.

Wones, D. R., 1981. Mafic silicates as indicators of intensive variables in granitic magmas. *Mining Geology* **31,** 191–212.

Wones, D. R., 1989. Significance of the assemblage titanite + magnetite + quartz in granitic rocks. *American Mineralogist* **74,** 744–749.

APPENDIX A: ANALYTICAL METHODS

The zircons were fractionated on the basis of their magnetic susceptibility (Table 1). For each of the magnetic fractions [m(1) and m(2)] about 100 large and 100 small crystals were picked. About 100 crystals were also picked from the non-magnetic fraction [nm(0)], but were not split on the basis of size. In all fractions, the crystals were selected to be as clear and free from fractures and inclusions as possible (Fig. A1). All fractions were

abraded until crystal faces were removed. Following abrasion, the zircon fractions, which weighed from 1·6 mg to ~240 μg, were dissolved in Teflon microcapsules by exposure to HF vapor. After dissolution, U and Pb were separated by ion exchange methods on microcolumns. Isotope ratio data were obtained with a VG Sector 54 multicollector instrument equipped with a Daly electron multiplier. Analytical data are given in Table 1 and plotted in Fig. 3. Data reduction and error analysis were done with the programs of Ludwig (1980, 1983); decay constants are those of Steiger & Jaeger (1977).

Minerals were analyzed on an automated four-spectrometer JEOL JXA-733 microprobe at Southern Methodist University. Accelerating voltage was 15 kV, beam current was 20 nA, and beam diameter was 20 μm. Standards consisted of a suite of natural and synthetic minerals; a modified Bence & Albee (1968) data reduction procedure was used.

Unexsolved feldspars were analyzed directly. The compositions of perthitic alkali feldspar were estimated with the use of a mass-balance scheme (Shannon, 1994) in which analyzed mineral compositions were summed to the whole-rock composition. This calculation yielded the volume proportions of perthite lamellae, which were then summed to give the integrated alkali feldspar composition.

Bulk samples for chemical analysis (2–4 kg) were unaltered and free of veins and weathered surfaces. They were crushed in steel jaw crushers and splits were powdered in a tungsten carbide shatterbox for inductively coupled plasma atomic emission spectrometry (ICP-AES) analysis, and in an alumina shatterbox for instrumental neutron activation analysis (INAA). The major elements, Sr, Zr, Y, Nb, Ba, Sc, V, Cu, Zn, and REE were analyzed by ICP-AES (see Shannon, 1994). For some samples, the REE, Ta, Th, Hf, and U were measured by INAA at Oregon State University. Rb was determined by flame emission, FeO by titration, and volatile content by loss on ignition. Relative uncertainties at the 99% confidence level (Student's *t* multiplied by standard deviation divided by the mean value for repeated analyses) are <1% for Si and Al; <2% for V; <3% for Ti, Fe, Mn, Ca, Na, La, Sr, Ba, and Zr; <5% for K, Mg, Y, Rb, Cs, Hf, Ta, Th, Sm, Eu, Tb, Yb, and Lu; <10% for P, Ce, U, Sc, Cu, and Zn; <15% for Nb, Nd, Cr, and Ni.